**ADITYA UNIVERSITY**

(Formerly Aditya Engineering College (A))

## Project-PO Mapping

**A.Y:** 2025-26 (AR 20) **Sem.:** VIII **Section & Batch: A2**

Project Title: **ANDROID BASED PICK AND PLACE ROBOT VEHICLE**


## Abstract

This project presents the design and development of a mobile industrial pick-and-place robotic arm with a soft gripper for safe object handling. The system integrates mechanical design, embedded control, and wireless communication to perform tasks such as material handling, assembly, inspection, and pick-and-place operations with improved efficiency and precision [PO1, PO2].

The robotic arm has 3 Degrees of Freedom (DOF) and is mounted on a mo

bile platform capable of moving in forward, backward, left, and right directions. Geared motors and a gripper mechanism ensure accurate motion and safe manipulation of objects, addressing practical industrial and domestic requirements [PO3, PO6].

The system uses a microcontroller-based control unit with an L293D motor driver, managing multiple motors for vehicle movement, arm lifting, and gripper operation. Wireless control is achieved through an Android application using Bluetooth, with additional support for voice commands, providing a user-friendly and flexible interface [PO4, PO5]. Performance validation is carried out by testing motion control, gripping accuracy, communication reliability, and response time under different operating conditions, ensuring dependable system functionality [PO4, PO5, PO7].

This project emphasizes low-cost automation, ethical considerations, and the impact of robotics in industrial, domestic, and hazardous environments, where it can reduce human effort and improve safety in repetitive or risky tasks [PO8, PO9, PO11].

The results demonstrate a reliable, efficient, and scalable robotic solution suitable for modern automation needs [PO3, PO12].



**Keywords**: Embedded System, Bluetooth, Robotic ARM, Arduino


| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Overall Mapping | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

| | PSO1 | PSO2 |
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| Overall Mapping | 1 | 2 |

**Signature of Project Members:**

**Guide Signature**

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# CHAPTER 1

# INTRODUCTION

## 1 Introduction to Robotics

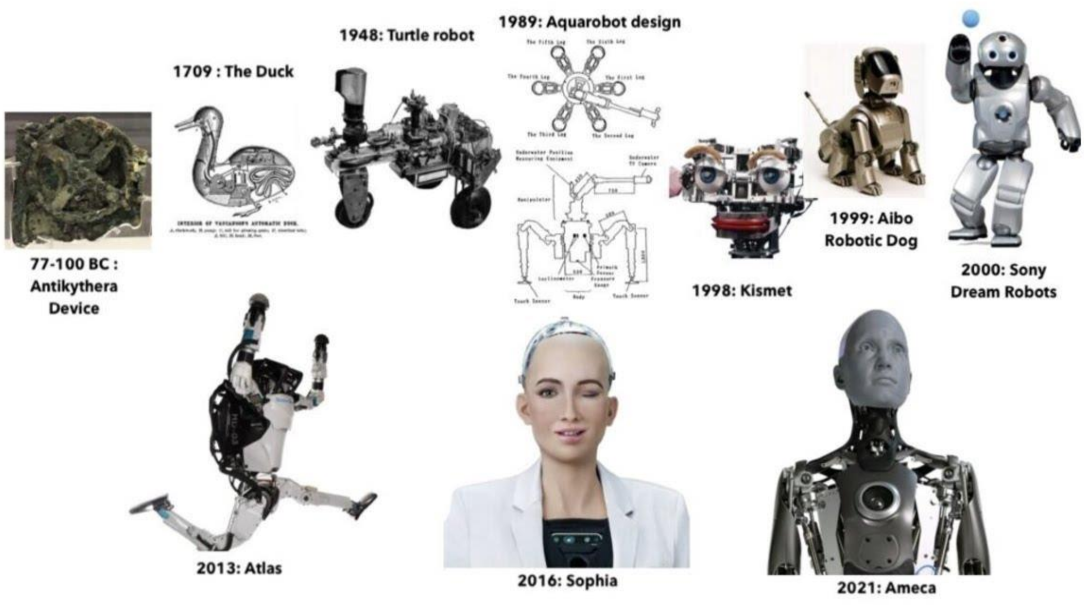
Robotics is a multidisciplinary field of engineering and science that deals with the design, construction, operation and application of robots. It integrates concepts from electronics, mechanical engineering, computer science, control systems and artificial intelligence to develop machines capable of performing tasks automatically or semi-automatically. With rapid advancements in sensors, actuators, embedded controllers and communication technologies, robotics has become a key component in modern industries, healthcare, agriculture, defence and even domestic environments. Robots are increasingly used to handle tasks that are difficult, repetitive, time-consuming or hazardous for humans, thereby improving overall productivity and safety.

## 1.1 Definition of Robot and Applications

A robot can be defined as a programmable electro-mechanical machine that is capable of carrying out a sequence of actions with precision and repeatability. It typically consists of a mechanical structure, actuators, sensors, a controller and a suitable power supply. Robots are especially useful in environments that are dangerous, highly repetitive or require high accuracy, where human operators may become fatigued or prone to error. Depending on their level of autonomy, robots may be fully autonomous, following pre-programmed instructions and sensor feedback, or manually controlled through wired or wireless interfaces such as joysticks, computers or mobile applications. Common application areas include industrial automation, material handling, welding, painting, medical surgery, military reconnaissance, domestic cleaning and educational training**.**

### The Evolution of Robots

**Fig 1:** Evolution of robots



The evolution of robots has spanned from ancient automated myths to modern AI-driven systems. Key milestones include 1950s industrial arms like Unimate, the rise of mobile robots in the 1960s, and 21st-century humanoid companions and cobots. Today, robots are increasingly autonomous, using AI and sensors to work alongside humans in manufacturing, medicine, and daily life.

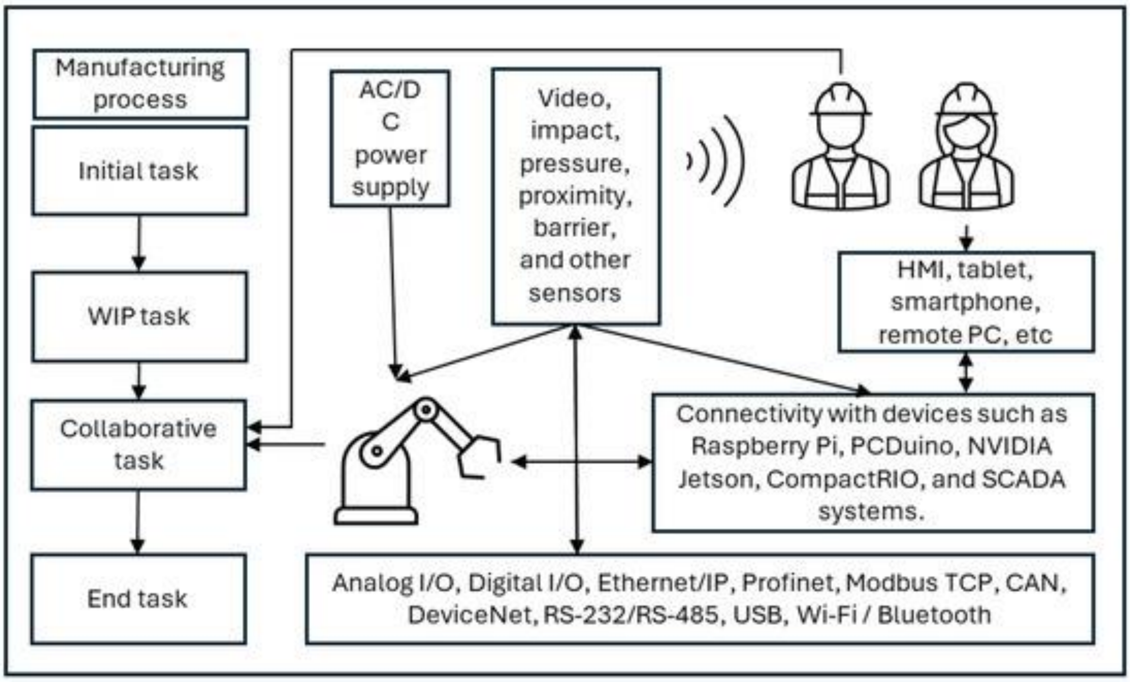



**Fig 2:** Industrial Robotic Arms

**Key Eras of Robotic Evolution:**

- **Ancient & Classical Origins:** Early automatons and mechanical creatures created for entertainment or ritualistic purposes, such as Heron of Alexandria's steam-powered devices or mythical stone giants.

- **Industrial Revolution:** Development of motors and power, leading to early automated mechanisms.

- **1950s–1970s (The Birth of Modern Robotics):**
  - **1954:** George Devol designed the first industrial robot, Unimate, later used by General Motors for, welding and lifting.
  - **1966:** Shakey the Robot became the first mobile robot that could perceive its environment.
  - **1973:** Waseda University developed the first full-scale anthropomorphic robot, Wabot-1.
- **1980s–1990s (Increased Capability):** Robots developed finer motor skills and advanced sensors. Honda began developing its P-series humanoids, leading to the advanced P2 and P3 models that could climb stairs and walk.

- **2000s–Present (AI, Intelligence, and Collaboration):**
  - **2000s:** Introduction of social robots like Sony's QRIO and Roomba vacuum cleaners.
  - **2010s–2020s:** The rise of collaborative robots (cobots) that work safely alongside humans.
  - **Modern Era:** Integration of advanced artificial intelligence (AI), machine vision, and machine learning, allowing robots to adapt, learn, and perform complex tasks, including space exploration (Mars Perseverance rover) and service-oriented humanoid roles.

### Pick and Place Robots

One of the most common and important applications of robotics is the pick and place robot. A pick and place robot is designed to pick up objects from one location and place them in another. This type of robot is widely used in manufacturing industries, packaging units, warehouses, and assembly lines. The basic components of a pick and place robot include a robotic arm or gripper mechanism, actuators (such as motors), a controller (like a microcontroller or PLC), and sensors for feedback and positioning. The working principle of a pick and place robot involves identifying an object, moving towards it, gripping it using a mechanical gripper or suction mechanism, and placing it at a desired location. These operations can be controlled manually or automated using pre-programmed instructions, and advanced systems may include sensors and vision systems for accurate object detection and placement.

### 1.2 Problem Statement

In many small and medium-scale industries, pick and place operations are still performed manually or with expensive fully automated robotic systems. Manual handling leads to fatigue, reduced efficiency, and higher chances of human error, while high-end industrial robots are costly, complex to program, and not easily accessible for educational or prototype applications. There is a need for a low-cost, user-friendly, and portable pick and place robotic system that can be controlled wirelessly and can demonstrate industrial concepts on a smaller scale. This project addresses this problem by developing an Android-based pick and place robot suitable for laboratory and light industrial use.

### 1.3 Motivation and Scope

The motivation behind this project is to design an affordable and practical robotic system that helps students and small industries understand and implement automation. Recent advances in low-cost microcontrollers like Arduino, along with wireless technologies such as Bluetooth and Wi-Fi, make it possible to build efficient pick and place robots using easily available components. The scope of this work includes the design and development of a Bluetooth-controlled pick and place robot capable of moving in multiple directions, gripping objects, and transporting them over short distances, making it useful for educational demonstrations, mini-industrial setups, and prototype material handling applications.

## 1.4 Organization of Report

This report is organized as follows. Chapter 1 introduces robotics, defines robots, and explains the concept of pick and place robots along with the problem statement and motivation. Chapter 2 presents the literature survey of existing robotic systems related to pick and place applications. Chapter 3 explains the system analysis, including existing and proposed system details. Chapter 4 describes the hardware and software design of the Android-based pick and place robot. Chapter 5 covers the implementation, testing, and integration of the system. Chapter 6 discusses the results and observations, and Chapter 7 presents the conclusion and future scope of the project.

## 1.5 Objective

- To design and develop a wireless pick and place robot using Arduino.
- To control the robot using an Android mobile application via Bluetooth.
- To implement real-time status display using an LCD module.
- To enable picking, storing, and placing of small objects efficiently.
- To understand integration of hardware components like motor driver, Bluetooth module, and microcontroller.

# CHAPTER 2

# LITERATURE SURVEY

**Table 1:** Literature Review

| S. No. | Author / Year | Title of Work | Technology Used | Description |
|---|---|---|---|---|
| 1 | Smith et al. (2018) | Bluetooth Controlled Robotic Arm | Arduino, HC-05, DC Motors | Developed a robotic arm controlled via Bluetooth mobile app for basic object handling. |
| 2 | Kumar & Rao (2019) | Pick and Place Robot Using Microcontroller | Microcontroller, Motor Driver | Designed a basic pick and place robot for industrial applications. |
| 3 | Lee et al. (2020) | Wireless Robotic Control System | Wi-Fi Module, Embedded System | Implemented wireless robot control using mobile devices for industrial automation. |
| 4 | Sharma et al. (2021) | Arduino Based Material Handling Robot | Arduino Uno, L298 Motor Driver | Developed a robot for transporting small materials using DC motors. |
| 5 | Patel & Singh (2022) | Mobile Controlled Robot with LCD Feedback | Arduino, Bluetooth, LCD | Integrated LCD display to show robot status and commands. |
| 6 | Ahmed et al. (2023) | Smart Pick and Place Robot | Arduino, Sensors, Gripper Mechanism | Designed a robot with gripping mechanism for picking objects automatically. |
| 7 | Reddy et al. (2024) | Android Controlled Robotic Vehicle | Arduino, HC-05, Mobile App | Developed a mobile-controlled robotic vehicle using Bluetooth communication. |

[1]. John Iovino [1], Various topics of robot design are discussed in this paper. It covers a range of issues such as arm design, control techniques, and vehicle design. The functioning and control of robots is discussed in this publication.

[2]. Akratic Kaushik, Aastha Sharma [2], Sensors provide a means of acquiring information about industrial activities and processes, as discussed in this paper. Sensors are frequently employed to convert a physical stimulus into an electronic signal that the manufacturing system can analyse and utilise to make decisions about the processes being executed.

[3].S. Senthilkumar1, L. Ramachandran. S. Aarthi [3],This paper proposes a system that uses an IR sensor in the designed pick and place robot to alert the user if an objects available in front of the vehicle.

[4]. These systems often employ hardware platforms like Raspberry Pi, Arduino, or Jetson Nano, coupled with efficient motion planning algorithms and real-time feedback mechanisms. Applications range from sorting and assembly tasks in industrial environments to educational tools for introducing STEM concepts. The studies emphasize adaptability, cost-effectiveness, and real-time performance as key strengths. Innovations like IOT integration, gesture-based control, and deep learning have broadened the scope of robotic arms, enabling their use in automation, manufacturing, and logistics.

[5]. Kumar Aaditya et al. (2015) demonstrated an Android-controlled, microcontroller-driven pick-and-place robot with GUI-based remote operation, providing a safer alternative for hazardous and critical industrial sites. Collectively, these works illustrate a consistent trend toward smarter, safer, and more versatile robotic automation through embedded systems, wireless connectivity, and intelligent control mechanisms. **-** IRJET (journal).

[6]. Vijay Matta et al. (2018) and Kaustubh Ghadge et al. (2018) both emphasized cost-effectiveness and reprogram ability, designing compact robotic systems with wireless or Wi-Fi control, suitable for both educational and industrial application.

[7]. N. U. Alka et al. (2017), who engineered a voice-controlled pick-and-place arm leveraging an ATMEGA328P and Android app, demonstrating practical solutions for remote and hazardous task execution. – AJER (journal)

[8]. Nita M. Thakre and team (2025) introduced a programmable robotic arm integrating RGB sensors and computer vision with an Android interface, enabling precise and real-time object sorting, thus addressing rigidity and detection limitations of traditional arms.

### 2.1 Summary of Existing Works

Several researchers have developed robotic systems for pick-and-place and mobile handling tasks using microcontrollers and wireless control. Smith et al. (2018) implemented a Bluetooth-controlled robotic arm using Arduino, HC-05 module, and DC motors to perform basic object handling via a mobile application. Kumar and Rao (2019) designed a microcontroller-based pick and place robot focused on industrial material transfer using a motor driver for actuation.

Lee et al. (2020) proposed a wireless robotic control system using a Wi-Fi module and embedded controller to achieve long-range monitoring and control for industrial automation. Sharma et al. (2021) developed an Arduino Uno and L298-based material handling robot for transporting small materials in a controlled environment. Patel and Singh (2022) integrated an LCD display into a mobile-controlled robot to provide real-time status feedback of commands and robot actions. Ahmed et al. (2023) introduced a smart pick and place robot with sensors and a gripper mechanism for semi-automatic object picking. Reddy et al. (2024) presented an Android-controlled robotic vehicle using Arduino and HC-05 for wireless motion control.

### 2.2 Analysis of Previous Systems

From the reviewed works, it is clear that Arduino-based platforms with Bluetooth or Wi-Fi communication are widely used due to their low cost, flexibility, and ease of programming. However, many systems focus either only on robotic arm manipulation or only on vehicle motion, and do not always combine a mobile platform with a pick-and-place mechanism plus user feedback. Some designs lack portability for real industrial environments or are primarily meant for laboratory demonstrations, while others require relatively complex hardware or programming setups.

These observations highlight a gap for a system that is simple, low-cost, and portable, yet still demonstrates core industrial features such as multi-directional movement, pick-and-place operation, and status indication. The proposed Android-based pick and place robot addresses this by using Arduino, HC-05 Bluetooth, motor drivers, and an LCD to provide wireless control, mobility, gripper operation, and real-time feedback in a single integrated platform suitable for education and small-scale industrial use**.**

# CHAPTER 3

# SYSTEM ANALYSIS

## 3.1 Existing System

In traditional systems, pick and place operations are either performed manually or by using fully automated industrial robotic arms. Manual systems require human effort, which leads to fatigue, reduced efficiency, and increased chances of errors. Moreover, manual handling can be unsafe in hazardous environments.

Industrial robotic systems, on the other hand, are highly efficient but come with several limitations:

- High cost of installation and maintenance
- Complex programming and setup
- Lack of portability
- Limited accessibility for small-scale industries and educational use
- Wired control systems reduce flexibility and mobility

Basic robotic systems without wireless control require direct connections, making them inconvenient to operate and limiting their range of movement

## 3.2 Limitations of Existing System

1. Manual pick and place operations require continuous human effort, which leads to fatigue and reduced efficiency.
2. Manual handling increases the chances of human errors during operation.
3. Manual handling can be unsafe in hazardous or risky environments.
4. Fully automated industrial robotic systems have high installation and maintenance cost.
5. Industrial robots often need complex programming and setup, making them difficult to use. Many industrial robotic systems lack portability.
6. They have limited accessibility for small-scale industries and educational institutions.
7. Wired control systems reduce flexibility and mobility of the robot.
8. Basic robotic systems without wireless control require direct connections, making them inconvenient to operate and limiting their range of movement.

### 3.3 PROPOSED SYSTEM

The proposed system introduces a Bluetooth / Wi-Fi-controlled pick and place robot that overcomes the limitations of existing systems by providing a low-cost, flexible and user-friendly solution. It is designed to reduce manual effort, improve safety and demonstrate industrial automation concepts in a compact form suitable for laboratories and small-scale setups.

In this system, an Arduino Uno is used as the main controller, which processes commands sent from an Android mobile application through Bluetooth or an ESP32 Wi-Fi module. The ESP32 establishes a wireless link between the mobile device and the robot, and the received commands are decoded by the Arduino to control the motion of the vehicle and the pick-and-place arm. The L298 motor driver is used to drive the DC geared motors based on these control signals, enabling forward, reverse, left, right movements and gripper operation.

Key features of the proposed system include wireless control using an Android mobile app over Wi-Fi (or Bluetooth, if required), eliminating the need for wired connections and giving the user freedom to operate the robot from a safe distance. The mobile platform provides mobility so that the robot can move in multiple directions and transport objects from one place to another within its working area. A dedicated pick and place mechanism with a gripper is used to grip and release small objects efficiently, while a small container at the rear side is provided for temporary storage and transport of picked items.

For user feedback, a 16×2 LCD module is interfaced with the controller to display real-time status messages such as connection status, received commands and current operations. The entire system is powered by a 12 V battery, which makes the robot portable and independent of a fixed power outlet. A transformer-based charging unit is provided so that the battery can be recharged safely when required, without removing it from the system.

With this configuration, the proposed Bluetooth or Wi-Fi-controlled pick and place robot is well suited for educational demonstrations, small-scale automation and prototype development, helping students and beginners gain hands-on experience in embedded systems, motor control, wireless communication and robotics.

### 3.4 Features of Proposed System

1. Wireless control using an Android mobile application via Bluetooth (HC-05), removing the need for wired connections.
2. Mobility of the robot platform, allowing movement in multiple directions and transport of objects.
3. Pick and place mechanism with a gripper that can efficiently grip and release small objects.
4. Material storage using a container at the back to store and transport picked items.
5. User feedback through an LCD module that displays real-time status messages, commands, and actions.
6. Portable power supply using a 12 V battery, making the robot independent of mains power.
7. Rechargeable system with a transformer-based charging circuit for safe battery recharging.
8. Overall low-cost, flexible, and user-friendly design suitable for education, small-scale automation, and prototype development.

### 3.5 Applications of Pick and Place Robot

1. Used for surveillance and for picking up harmful objects like bombs and diffusing them safely.
2. Used in manufacturing to pick required parts and place them in the correct position to complete fixtures, to load objects onto conveyor belts, and to remove defective products from the conveyor.
3. Used in various surgical operations such as joint replacement, orthopedic procedures, and internal surgery, where high precision and accuracy are required.

### 3.6 Advantages of Pick and Place Robot

1. They are faster and can complete tasks in seconds compared to human operators.
2. They are flexible and have an appropriate design for different applications.
3. They are accurate in positioning and handling objects.
4. They increase the safety of the working environment and never get tired**.**

### 3.7 Technical Description

- **Number of axes** – two axes are required to reach any point in a plane; three axes are required to reach any point in space. To fully control the orientation of the end of the arm (i.e. the wrist) three more axes (yaw, pitch, and roll) are required. Some designs (e.g. the SCARA robot) trade limitations in motion possibilities for cost, speed, and accuracy.
- Degrees of freedom – this is usually the same as the number of axes.
- Working envelope – the region of space a robot can reach.
- Kinematics – the actual arrangement of rigid members and joints in the robot, which determines the robot's possible motions. Classes of robot kinematics include articulated, cartesian, parallel and SCARA.
- **Carrying capacity or** payload – how much weight a robot can lift.
- **Speed** – how fast the robot can position the end of its arm. This may be defined in terms of the angular or linear speed of each axis or as a compound speed i.e. the speed of the end of the arm when all axes are moving.
- **Acceleration** – how quickly an axis can accelerate. Since this is a limiting factor a robot may not be able to reach its specified maximum speed for movements over a short distance or a complex path requiring frequent changes of direction.
- **Accuracy** – how closely a robot can reach a commanded position. When the absolute position of the robot is measured and compared to the commanded position the error is a

measure of accuracy. Accuracy can be improved with external sensing for example a vision system or Infra-Red. See robot calibration. Accuracy can vary with speed and position within the working envelope and with payload (see compliance).

- **Repeatability** – how well the robot will return to a programmed position. This is not the same as accuracy. It may be that when told to go to a certain X-Y-Z position that it gets only to within 1 mm of that position. This would be its accuracy which may be improved by calibration. But if that position is taught into controller memory and each time it is sent there it returns to within 0.1mm of the taught position then the repeatability will be within 0.1mm.

- **Motion control** – for some applications, such as simple pick-and-place assembly, the robot need merely return repeatably to a limited number of pre-taught positions. For more sophisticated applications, such as welding and finishing (spray painting), motion must be continuously controlled to follow a path in space, with controlled orientation and velocity.

- **Power source** – some robots use electric motors, others use hydraulic actuators. The former are faster, the latter are stronger and advantageous in applications such as spray painting, where a spark could set off an explosion; however, low internal air-pressurisation of the arm can prevent ingress of flammable vapours as well as other contaminants.

- **Drive** – some robots connect electric motors to the joints via gears; others connect the motor to the joint directly (direct drive). Using gears results in measurable 'backlash' which is free movement in an axis. Smaller robot arms frequently employ high speed, low torque DC motors, which generally require high gearing ratios; this has the disadvantage of backlash. In such cases the harmonic drive is often used.

- **Compliance** - this is a measure of the amount in angle or distance that a robot axis will move when a force is applied to it. Because of compliance when a robot goes to a position carrying its maximum payload it will be at a position slightly lower than when it is carrying no payload. Compliance can also be responsible for overshoot when carrying high payloads in which case acceleration would need to be reduced.

**End-of-arm tooling**

The most essential robot peripheral is the end effectors, or end-of-arm-tooling (EOT). Common examples of end effectors include welding devices (such as MIG-welding guns, spot-welders, etc.), spray guns and also grinding and deburring devices (such as pneumatic disk or belt grinders, burrs, etc.), and grippers (devices that can grasp an object, usually electromechanical or pneumatic). Another common means of picking up an object is by vacuum. End effectors are frequently highly complex, made to match the handled product and often capable of picking up an array of products at one time. They may utilize various sensors to aid the robot system in locating, handling, and positioning products.

# CHAPTER 4

# System Design

## 4 Hardware and Software Requirements

### Hardware Requirements

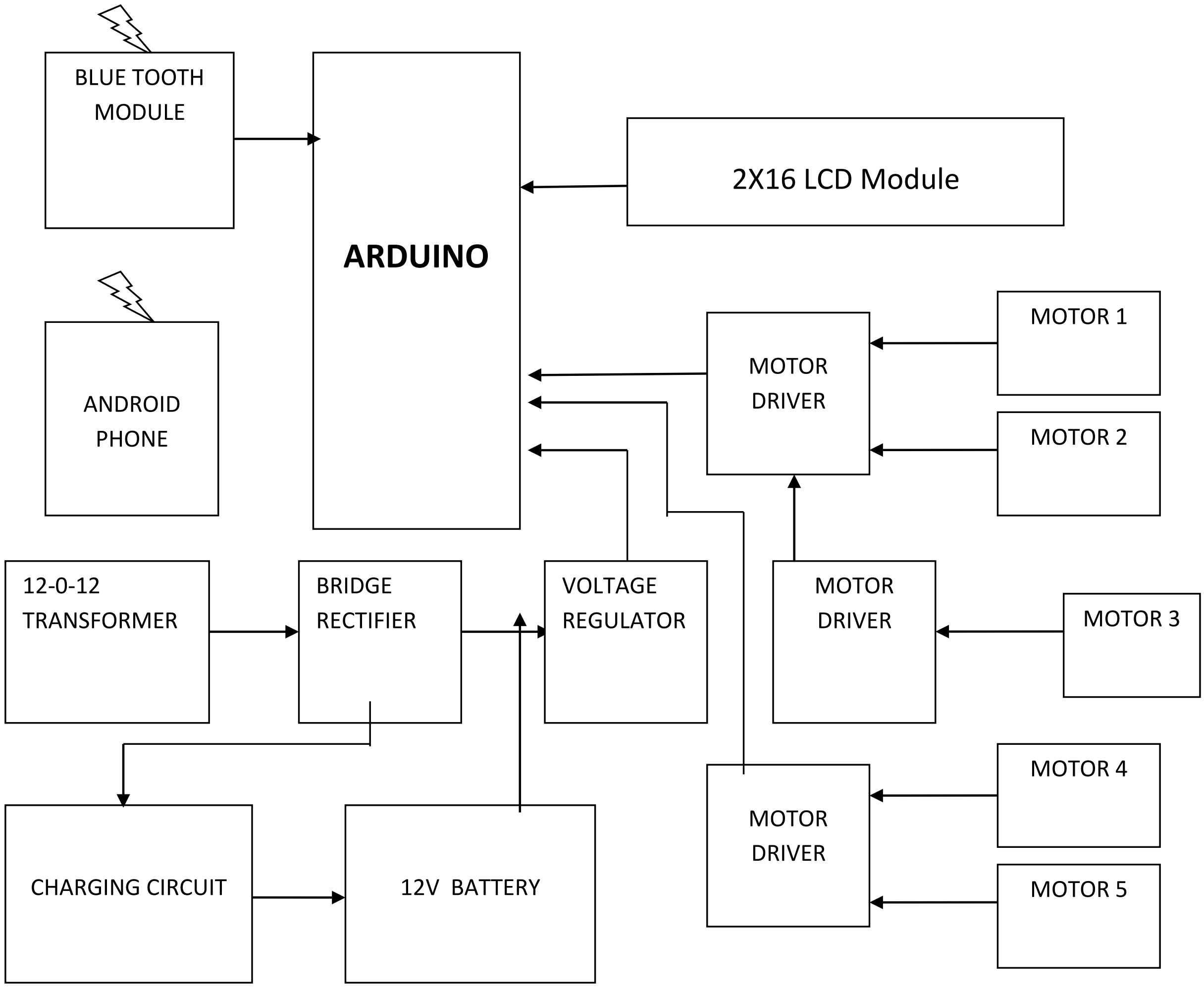
- Arduino UNO Controller
- LCD 2 x 16 Module
- L293D Motor Driver
- L298 Motor Driver
- 12V / 2.2 Ah Battery
- Bluetooth Module
- DC Geared Motor
- Android Application
- Chassis
- Clamps and Tyres
- Voltage Regulators
- Filter Capacitors
- Other Misc. Components

### Software Requirements

- Embedded C Programming Language
- Arduino IDE Compiler

## 4.1

## BLOCK DIAGRAM




## 4.2 Working / Methodology

The basic function of a pick and place robot is done by its joints. Joints are analogous to human joints and are used to join the two consecutive rigid bodies in the robot. They can be rotary joint or linear joint. To add a joint to any link of a robot, we need to know about the degrees of freedom and degrees of movement for that body part. Degrees of freedom implement the linear and rotational movement of the body and Degrees of movement imply the number of axis the body can move. A simple pick and place robot consists of two rigid bodies on a moving base, connected together with rotary joint.

In this Project we have designed a 3 Degree Freedom Industrial ARM. These types of Industrial ARM are mostly used in Automobile Industries, Packaging Industries, Automatic

Welding and Painting Industries. Movement of this ARM include Sideways – Left / Right, Up / Down, and Gripper Control – Open / Close. Atmega 16 Microcontroller is used as the Brain of this Industrial ARM. Android Application is used to control the ARM for various movements.

We can control this ARM using Mobile Phone / Tablet that has Android Operating System. Bluetooth Technology is used to communicate with Mobile phone and the ARM. This Android Application is designed using **MIT App inventor** Online web platform. The android application device transmitter acts as a remote control that has the advantage of adequate range, while the receiver end Bluetooth device is fed to the microcontroller to drive DC motors via motor driver IC for necessary work. User first connects the Android Application with the ARM. After successful connection a message “Bluetooth Connected” message will be displayed in the LCD module of the ARM. Now user can access the ARM using the Application controls (UP/DOWN/LEFT/RIGHT/OPEN/CLOSE). Robotic Vehicle can be moved in Forward, backward, Left and right Direction. To Disconnect the Application from the device, user has to press “Disconnect”. Arduino based Controller is used to design this system.

This entire System is powered using **12V Lithium battery**. The power supply and Charging section of the system contains a step-down transformer of 230/12V, used to step down the voltage to 12VAC. To convert it to DC, a bridge rectifier is used. Capacitive filter is used which makes use of 7805 voltage regulator to regulate it to +5V that will be needed for microcontroller and other components operation, in order to remove ripple.

### 4.3 Hardware Components Description

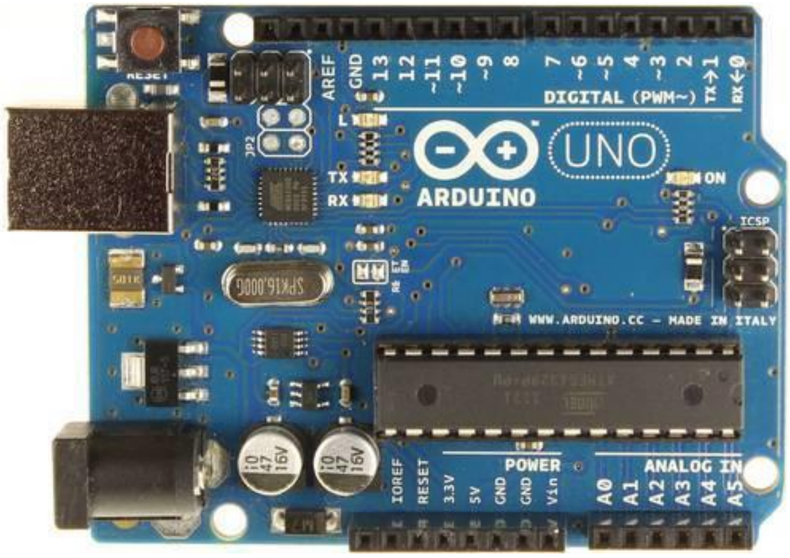
#### 4.3.1 ARDUINO UNO

**Arduino** is common term for a software company, project, and user community that designs and manufactures computer open-source hardware, open-source software, and microcontroller-based kits for building digital devices and interactive objects that can sense and control physical devices.[1]

The project is based on microcontroller board designs, produced by several vendors, using various microcontrollers. These systems provide sets of digital and analog I/O pins that can interface to various expansion boards (termed *shields*) and other circuits. The boards feature serial communication interfaces, including Universal Serial Bus (USB) on some models, for loading programs from personal computers. For programming the microcontrollers, the Arduino project provides an integrated development environment (IDE) based on a programming language named *Processing*, which also supports the languages C and C++.

The first Arduino was introduced in 2005, aiming to provide a low cost, easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion detectors.

**Fig 3**: Arduino UNO




Arduino programs may be written in any programming language with a compiler that produces binary machine code. Atmel provides a development environment for their microcontrollers, AVR Studio and the newer Atmel Studio.[19][20]

The Arduino project provides the Arduino integrated development environment (IDE), which is a cross-platform application written in the programming language Java. It originated from the IDE for the languages *Processing* and *Wiring*. It is designed to introduce programming to artists and other newcomers unfamiliar with software development. It includes a code editor with features such as syntax highlighting, brace matching, and automatic indentation, and provides simple one-click mechanism to compile and load programs to an Arduino board. A program written with the IDE for Arduino is called a "sketch".[21]

The Arduino IDE supports the languages C and C++ using special rules to organize code. The Arduino IDE supplies a software library called Wiring from the Wiring project, which provides many common input and output procedures. A typical Arduino C/C++ sketch consist of two functions that are compiled and linked with a program stub *main()* into an executable cyclic executive program:

- *setup()*: a function that runs once at the start of a program and that can initialize settings.
- *loop()*: a function called repeatedly until the board powers off.

After compiling and linking with the GNU toolchain, also included with the IDE distribution, the Arduino IDE employs the program *avrdude* to convert the executable code into a text file in hexadecimal coding that is loaded into the Arduino board by a loader program in the board's firmware.

**Applications**

See also: List of open-source hardware projects

- Xoscillo, an open-source oscilloscope
- Scientific equipment[27] such as the Chemduino
- Arduinome, a MIDI controller device that mimics the Monome
- OBDuino, a trip computer that uses the on-board diagnostics interface found in most modern cars
- Ard pilot, drone software and hardware
- Arduino Phone, a do-it-yourself cell phone
- *Gert Duino*, an Arduino mate for the Raspberry Pi
- Water quality testing platform
- Homemade CNC using Arduino and DC motors with close loop control by Homofaciens

**Arduino Uno – Details**

The Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

"Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0. The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases. The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform; for an extensive list of current, past or outdated boards see the Arduino index of boards.

**Table 2:** Technical Specs of Arduino

**Technical specs**

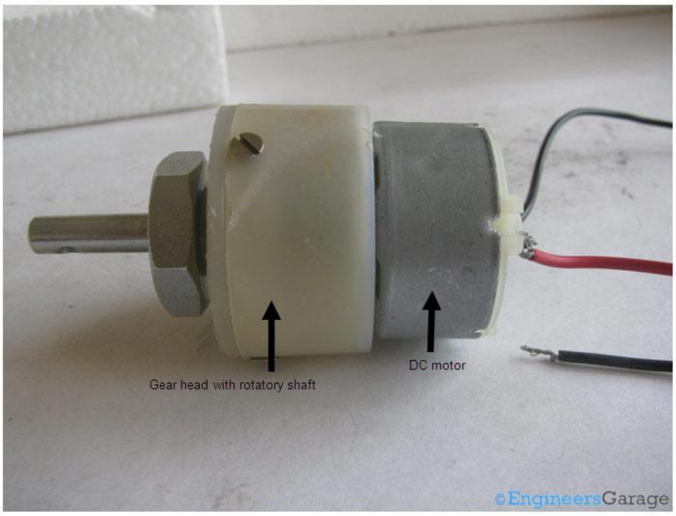
| | |
|---|---|
| Microcontroller | ATmega328P |
| Operating Voltage | 5V |
| Input Voltage (recommended) | 7-12V |
| Input Voltage (limit) | 6-20V |
| Digital I/O Pins | 14 (of which 6 provide PWM output) |
| PWM Digital I/O Pins | 6 |
| Analog Input Pins | 6 |
| DC Current per I/O Pin | 20 mA |
| DC Current for 3.3V Pin | 50 mA |
| Flash Memory | 32 KB (ATmega328P) |

| | |
|---|---|
| | of which 0.5 KB used by bootloader |
| SRAM | 2 KB (ATmega328P) |
| EEPROM | 1 KB (ATmega328P) |
| Clock Speed | 16 MHz |
| Length | 68.6 mm |
| Width | 53.4 mm |
| Weight | 25 g |

### 4.3.2 DC Geared Motor

Geared DC motors can be defined as an extension of DC motor which already had its Insight details demystified here. A geared DC Motor has a gear assembly attached to the motor. The speed of motor is counted in terms of rotations of the shaft per minute and is termed as RPM .The gear assembly helps in increasing the torque and reducing the speed. Using the correct combination of gears in a gear motor, its speed can be reduced to any desirable figure. This concept where gears reduce the speed of the vehicle but increase its torque is known as gear reduction.  This Insight will explore all the minor and major details that make the gear head and hence the working of geared DC motor.

**Fig 4:** DC Gear Motor




### Working of the DC Geared Motor

The DC motor works over a fair range of voltage. The higher the input voltage more is the RPM (rotations per minute) of the motor. For example, if the motor works in the range of 6-12V, it will have the least RPM at 6V and maximum at 12 V.

In terms of voltage, we can put the equation as:
RPM= K1 * V, where,
K1= induced voltage constant
V=voltage applied

The working of the gears is very interesting to know. It can be explained by the principle of conservation of angular momentum. The gear having smaller radius will cover more RPM than the one with larger radius. However, the larger gear will give more torque to the smaller gear than vice versa. The comparison of angular velocity between input gear (the one that transfers energy) to output gear gives the gear ratio. When multiple gears are connected together, conservation of energy is also followed. The direction in which the other gear rotates is always the opposite of the gear adjacent to it.

In any DC motor, RPM and torque are inversely proportional. Hence the gear having more torque will provide a lesser RPM and converse. In a geared DC motor, the concept of pulse width modulation is applied.

In a geared DC motor, the gear connecting the motor and the gear head is quite small, hence it transfers more speed to the larger teeth part of the gear head and makes it rotate. The larger part of the gear further turns the smaller duplex part. The small duplex part receives the torque but not the speed from its predecessor which it transfers to larger part of other gear and so on. The third gear's duplex part has more teeth than others and hence it transfers more torque to the gear that is connected to the shaft.

**Features of DC Gear Motors**

- **Gear materials:** Plastic or metal.
- **Motor types:** Wound-field, permanent-magnet, brushless, intermittent and continuous duty motors.
- **Brush-type and brushless motors:** The brushed motor gains torque from the power supplied to the motor using stationary magnets, commutation and rotating electrical magnets. Brushless motors use a soft magnetic core in the rotor or a permanent magnet, as well as stationary magnets in the housing.
- **Uncommitted motors:** Homopolar motors or ball bearing motors.
- **Connection types:** Shunt, series and compound connections.
- **Motor constants:** Kv and Km.
- **Speed control and reversibility:** Smoothly control a speed down to zero without power circuit switching, even after accelerating in the opposite direction.
- **Dynamic braking and regenerative braking:** Ideal for applications that require quick stops so you don't need a mechanical brake.
- **Magnet types:** Rare earth, ceramic or ferrite magnets.
- **Winding resistance:** Choose a motor that doesn't adversely affect the Km.
- **Gear ratios:** Several varieties available, such as 28:1 or 18:1.

- **Environment:** Motors are available for indoor or outdoor use.
- **Torque multiplication:** Generate a large force at a low speed.
- **Custom-built:** You can have a DC gear motor designed and manufactured to suit your size, power, torque and mounting needs.

The uses for DC gear motors are nearly unlimited. The following **applications** are some that you may recognize:

- Battery-operated toys
- Disk drives
- Steel rolling mills
- Paper machines
- Medical equipment
- Radio-controlled aircraft
- Automobiles
- Drive systems
- Positioning, industrial and consumer actuators
- Winches
- Robotics
- Mixers, from cake to concrete
- Cranes
- Power drills
- Washing machine knobs
- Electromechanical clocks

### 4.3.3 Diodes

A diode is an electronic component with two electrodes (connectors). It allows electricity to go through it only in one direction.

Diodes can be used to convert alternating current to direct current (Diode bridge). They are often used in power supplies and sometimes to decode amplitude modulation radio signals (like in a crystal radio). Light-emitting diodes (LEDs) are a type of diode that produce light. Today, the most common diodes are made from semiconductor materials such as silicon or sometimes germanium.

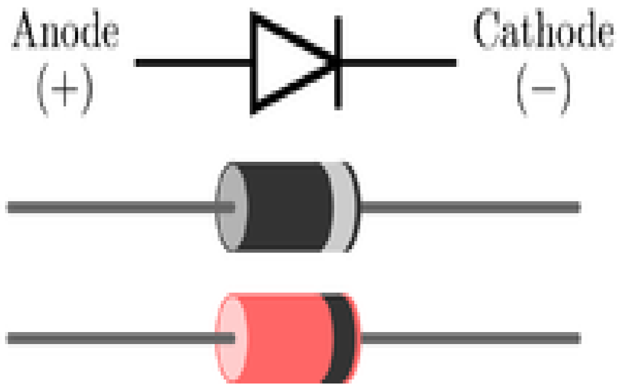
### Construction

Semiconductor diodes are made of two types of semiconductors connected to each other. One type has atoms with extra electrons (called the n-side). The other type has atoms that want electrons (called the p-side). Because of this, the electricity will flow easily from the side with too many electrons to the side with too few. However, electricity will not flow easily in the reverse direction. Silicon with arsenic dissolved in it makes a good n-side semiconductor, while silicon with aluminum dissolved in it makes a good p-side semiconductor, but other materials can also work.

The connector to the n-side is called the cathode, the connector to the p-side is called the anode.

**Function of a diode**

**Fig 5:** Diode Positive voltage at p-side.




If you give positive voltage to the p-side and negative voltage to the n-side, the electrons in the n-side will want to go to the positive voltage at the p-side and the holes of the p-side will want to go to the negative voltage at the n-side. Because of this, current flow is able to exist, but it takes a certain amount of voltage to get this started (very small amount of voltage is not enough to get the electric current to flow). This is called the cut-in voltage. The cut-in voltage of a silicon diode is at about 0.7 V. A germanium diode needs a cut-in voltage at about 0.3 V.

**Negative voltage at p-side**

If you instead give negative voltage to the p-side and positive voltage to the n-side, the electrons of the n-side want to go to the positive voltage source instead of the other side of the diode. Same thing happens on the p-side. So, current will not flow between the two sides of the diode. Increasing the voltage will eventually force electric current to flow (this is the break-down voltage). Many diodes will be destroyed by a reverse flow but some are made that can survive it.

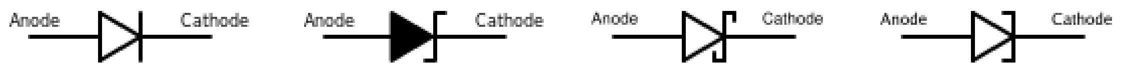
**Applications of Diodes**

1. Radio demodulation
2. Power conversion
3. Over-voltage protection
4. Logic gates
5. Temperature measuremen.

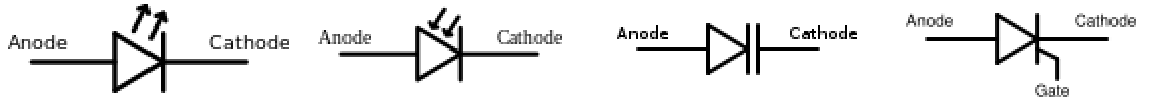
**Types of diodes**

Here are some common semiconductor diode symbols used in schematic diagrams:

**Fig 6:** Types of Diodes




Diode Zener Diode Schottky diode Tunnel diode




Light-emitting diode Photodiode Varicap Silicon controlled rectifier

**Functions of Diodes**

The most common function of a diode is to allow an electric current to pass in one direction (called the diode's *forward* direction), while blocking current in the opposite direction (the *reverse* direction). Thus, the diode can be viewed as an electronic version of a check valve. This unidirectional behaviour is called rectification, and is used to convert alternating current to direct current, including extraction of modulation from radio signals in radio receivers—these diodes are forms of rectifiers.

However, diodes can have more complicated behaviour than this simple on–off action, due to their nonlinear current-voltage characteristics. Semiconductor diodes begin conducting electricity only if a certain threshold voltage or cut-in voltage is present in the forward direction (a state in which the diode is said to be *forward-biased*). The voltage drop across a forward-biased diode varies only a little with the current, and is a function of temperature; this effect can be used as a temperature sensor or as a voltage reference.

A semiconductor diode's current–voltage characteristic can be tailored by selecting the semiconductor materials and the doping impurities introduced into the materials during manufacture. These techniques are used to create special-purpose diodes that perform many different functions. For example, diodes are used to regulate voltage (Zener diodes), to protect circuits from high voltage surges (avalanche diodes), to electronically tune radio and TV receivers (varactor diodes), to generate radio-frequency oscillations (tunnel diodes, Gunn diodes, IMPATT diodes), and to produce light (light-emitting diodes). Tunnel, Gunn and IMPATT diodes exhibit negative resistance, which is useful in microwave and switching circuits.

## 4.3.4 CAPACITORS

## Electrolyte Capacitor

An electrolytic capacitor is a type of capacitor that uses an electrolyte to achieve a larger capacitance than other capacitor types. An electrolyte is a liquid or gel containing a high concentration of ions. Almost all electrolytic capacitors are polarized, which means that the voltage on the positive terminal must always be greater than the voltage on the negative terminal. The benefit of large capacitance in electrolytic capacitors comes with several drawbacks as well. Among these drawbacks are large leakage currents, value tolerances, equivalent series resistance and a limited lifetime. Electrolytic capacitors can be either wet-electrolyte or solid polymer.


**Fig 7:** Electrolyte Capacitor

They are commonly made of tantalum or aluminium, although other materials may be used. Supercapacitors are a special subtype of electrolytic capacitors, also called double-layer electrolytic
capacitors, with capacitances of hundreds and thousands of farads. This article will be based on aluminium electrolytic capacitors.

These have a typical capacitance between 1µF to 47mF and an operating voltage of up to a few hundred volts DC. Aluminium electrolytic capacitors are found in many applications such as power supplies, computer motherboards and many domestic appliances. Since they are polarized, they may be used only in DC circuits.

## Characteristics

### Capacitance drift

The capacitance of electrolytic capacitors drifts from the nominal value as time passes, and they have large tolerances, typically 20%. This means that an aluminium electrolytic capacitor with a nominal capacitance of 47µF is expected to have a measured value of anywhere between 37.6µF and 56.4µF. Tantalum electrolytic capacitors can be made with tighter tolerances, but their maximum operating voltage is lower so they cannot be always used as a direct replacement.

**Polarity and safety**

Due to the construction of electrolytic capacitors and the characteristics of the electrolyte used, electrolytic capacitors must be forward biased. This means that the positive terminal must always be at a higher voltage than the negative terminal. If the capacitor becomes reverse-biased (if the voltage polarity on the terminals is reversed), the insulating aluminum oxide, which acts as a dielectric, might get damaged and start acting as a short circuit between the two capacitor terminals. This can cause the capacitor to overheat due to the large current running through it.

As the capacitor overheats, the electrolyte heats up and leaks or even vaporizes, causing the enclosure to burst. This process happens at reverse voltages of about 1 volt and above. To maintain safety and prevent the enclosure from exploding due to high pressures generated under overheat conditions, a safety valve is installed in the enclosure. It is typically made by making a score in the upper face of the capacitor, which pops open in a controlled manner when the capacitor overheats. Since electrolytes may be toxic or corrosive, additional safety measures may need to be taken when cleaning after and replacing an overheated electrolytic capacitor.

There is a special type of electrolytic capacitors for AC use, which is designed to withstand reverse polarisation. This type is called the non-polarized or NP type.

## Construction and properties of electrolytic capacitors

Aluminium electrolytic capacitors are made of two aluminium foils and a paper spacer soaked in electrolyte. One of the two aluminium foils is covered with an oxide layer, and that foil acts as the anode, while the uncoated one acts as a cathode. During normal operation, the anode must be at a positive voltage in relation to the cathode, which is why the cathode is most commonly marked with a minus sign along the body of the capacitor. The anode, electrolyte-soaked paper and cathode are stacked. The stack is rolled, placed into a cylindrical enclosure and connected to the circuit using pins. There are two common geometries: axial and radial. Axial capacitors have one pin on each end of the cylinder, while in the radial geometry, both pins are located on the same end of the cylinder.

Electrolytic capacitors have a larger capacitance than most other capacitor types, typically 1µF to 47mF. There is a special type of electrolytic capacitor, called a double-layer capacitor or a supercapacitor, whose capacitance can reach thousands of farads. The capacitance of an aluminium electrolytic capacitor is determined by several factors, such as the plate area and the thickness of the electrolyte. This means that a large capacitance capacitor is bulky and large in size.

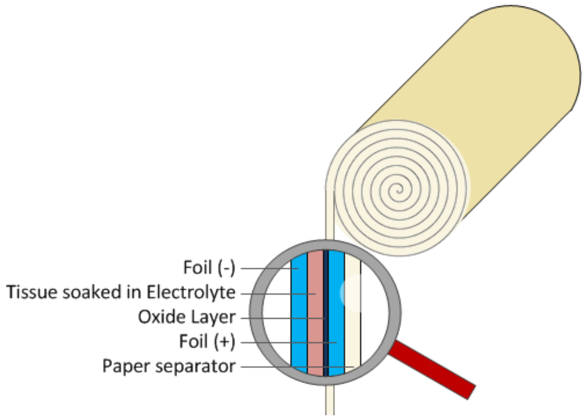



**Fig 8**: Aluminium Electrolyte Capacitor

It is worth mentioning that electrolytic capacitors made using old technology didn't have a very long shelf life, typically only a few months. If left unused, the oxide layer deteriorates and has to be rebuilt in a process called capacitor reforming. This can be performed by connecting the capacitor to a voltage source through a resistor and slowly increasing the voltage until the oxide layer has been fully rebuilt. Modern electrolytic capacitors have a shelf life of 2 years or more. If the capacitor is left unpolarized for longer periods, they must be reformed prior to use.

## Applications for electrolytic capacitors

There are many applications which do not need tight tolerances and AC polarization, but require large capacitance values. They are commonly used as filtering devices in various power supplies to reduce the voltage ripple. When used in switching power supplies, they are often the critical component limiting the usable life of the power supply, so high quality capacitors are used in this application.

They may also be used in input and output smoothing as a low pass filter if the signal is a DC signal with a weak AC component. However, electrolytic capacitors do not work well with large amplitude and high frequency signals due to the power dissipated at the parasitic internal resistance called equivalent series resistance (ESR). In such applications, low-ESR capacitors must be used to reduce losses and avoid overheating.

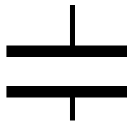
A practical example is the use of electrolytic capacitors as filters in audio amplifiers whose main goal is to reduce mains hum. Mains hum is a 50Hz or 60Hz electrical noise induced from the mains supply which would be audible if amplified.

### NON-ELECTROLYTIC CAPACITOR

Non-Electrolytic capacitors are non-polarised, i.e they can be connected either way in a circuit without having to worry about + & -. The most common is the disc-type capacitor that we normally use in electronics. The other types are ceramic, mica etc. In almost all applications we use the disc-type capacitor which is brown in colour and has the shape of a disc. Its value ranges between about a few pF to as high as 1uF. ( You also get non-polarised capacitors of higher values and such capacitors have 'NP' written on them indicating non-polarised)

The symbol for non-electrolytic capacitor is:

**Fig 9:** Capacitor Symbol



where the dark lines indicate the two plates, and the thin lines represent the two terminals.

**Non-Electrolytic:**
Some capacitors have their values printed in them. Unfortunately, there are various formats for printing the values and only a few can be discussed here:

1) If the printed value is like 101,102,103,204 etc then the value of the capacitor= (first 2 digits X 10 raised to the 3rd digit) pF.

For example, if the value is 104 then capacitance = ( 10X 10e4) pF = 10e5 pF = 10e-7 F = 0.1uF
Remember a few of them: 104 = 0.1uF, 224=0.22uF , 103= 0.01uF, 102= 0.001uF

2) If the printed value is like 1K5, 100,220,10K etc,
Then capacitance = (printed value) pF
For example if the value is 10K then capacitance = 10K pF = 10X10e3 pF= 10e-8 F= 0.01uF
1K5 means 1.5K pF and so on.

**Fig 10:** Non-Polarised Capacitor

A **ceramic capacitor** is a non-polarized fixed capacitor made out of two or more alternating layers of ceramic and metal in which the ceramic material acts as the dielectric and the metal acts as the electrodes. The ceramic material is a mixture of finely ground granules of paraelectric or ferroelectric materials, modified by mixed oxides that are necessary to achieve the capacitor's desired characteristics.

The electrical behaviour of the ceramic material is divided into two stability classes:

- Class 1 ceramic capacitors with high stability and low losses compensating the influence of temperature in resonant circuit application. Common EIA/IEC code abbreviations are C0G/NP0, P2G/N150, R2G/N220, U2J/N750 etc.
- Class 2 ceramic capacitors with high volumetric efficiency for buffer, by-pass and coupling applications Common EIA/IEC code abbreviations are: X7R/2XI, Z5U/E26, Y5V/2F4, X7S/2C1, etc.

The great plasticity of ceramic raw material works well for many special applications and enables an enormous diversity of styles, shapes and great dimensional spread of ceramic capacitors. The smallest discrete capacitor, for instance, is a "01005" chip capacitor with the dimension of only 0.4 mm × 0.2 mm.

The construction of ceramic multilayer capacitors with mostly alternating layers results in single capacitors connected in parallel. This configuration increases capacitance and decreases all losses and parasitic inductances. Ceramic capacitors are well-suited for high frequencies and high current pulse loads.

Because the thickness of the ceramic dielectric layer can be easily controlled and produced by the desired application voltage, ceramic capacitors are available with rated voltages up to the 30 kV range.

Some ceramic capacitors of special shapes and styles are used as capacitors for special applications, including RFI/EMI suppression capacitors for connection to supply mains, also known as safety capacitors, X2Y® and three-terminal capacitors for bypassing and decoupling applications, feed-through capacitors for noise suppression by low-pass filters[ and ceramic power capacitors for transmitters and HF applications.

**Film capacitors**

Film capacitors or plastic film capacitors are non-polarized capacitors with an insulating plastic film as the dielectric. The dielectric films are drawn to a thin layer, provided with metallic electrodes and wound into a cylindrical winding. The electrodes of film capacitors may be metallized aluminum or zinc, applied on one or both sides of the plastic film, resulting in metallized film capacitors or a separate metallic foil overlying the film, called film/foil capacitors.

Metallized film capacitors offer self-healing properties. Dielectric breakdowns or shorts between the electrodes do not destroy the component. The metallized construction makes it possible to produce wound capacitors with larger capacitance values (up to 100 µF and larger) in smaller cases than within film/foil construction.

Film/foil capacitors or metal foil capacitors use two plastic films as the dielectric. Each film is covered with a thin metal foil, mostly aluminium, to form the electrodes. The advantage of this construction is the ease of connecting the metal foil electrodes, along with an excellent current pulse strength.

A key advantage of every film capacitor's internal construction is direct contact to the electrodes on both ends of the winding. This contact keeps all current paths very short. The design behaves like a large number of individual capacitors connected in parallel, thus reducing the internal ohmic losses (ESR) and ESL. The inherent geometry of film capacitor structure results in low ohmic losses and a low parasitic inductance, which makes them suitable for applications with high surge currents (snubbers) and for AC power applications, or for applications at higher frequencies.

The plastic films used as the dielectric for film capacitors are Polypropylene (PP), Polyester (PET), Polyphenylene sulfide (PPS), Polyethylene naphthalate (PEN), and Polytetrafluoroethylene or Teflon (PTFE). Polypropylene film material with a market share of something about 50% and Polyester film with something about 40% are the most used film materials. The rest of something about 10% will be used by all other materials including PPS and paper with roughly 3%, each.

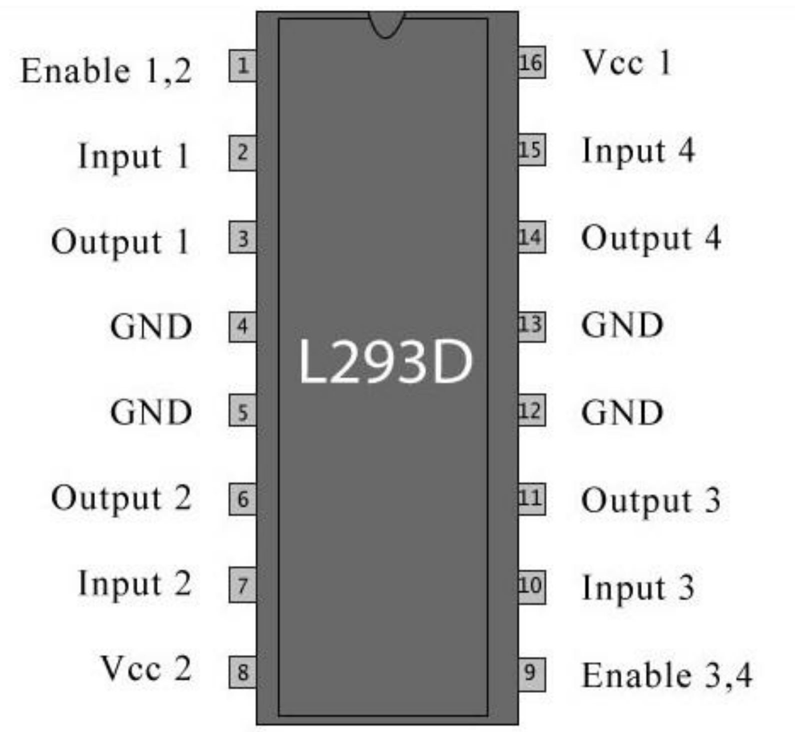



**Fig 11:** L293D Motor Driver

### 4.3.5 L293D Motor Driver IC

**L293D** is a typical Motor driver or Motor Driver integrated circuit which is used to drive direct current on either direction. It is a 16-pin IC which can control a set of two DC motors simultaneously in any direction. It means that you can control two DC motor with a single L293D IC. Dual H-bridge Motor Driver integrated circuit (IC). The L293D can drive small and quite big motors as well.

A **H bridge** is an electronic circuit that enables a voltage to be applied across a load in either direction. These circuits are often used in robotics and other applications to allow DC motors to run forwards and backwards.[1]

Most DC-to-AC converters (power inverters), most AC/AC converters, the DC-to-DC push–pull converter, most motor controllers, and many other kinds of power electronics use H bridges. In particular, a bipolar stepper motor is almost invariably driven by a motor controller containing two H bridges.

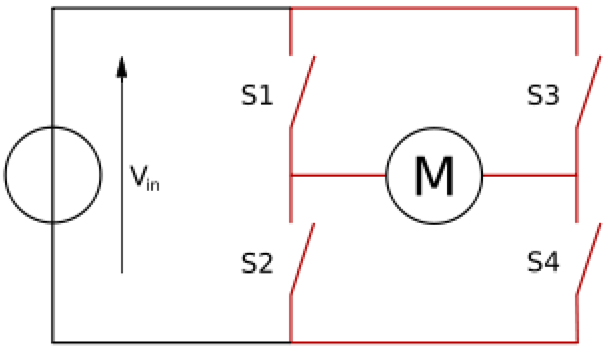



**Fig 12:** H-BRIDGE Concept

It works on the concept of H-bridge, a circuit which allows the high voltage to be flown in either direction. In a single L293D IC there two H-bridge circuit inside the it which can rotate two dc motor independently. Due to its size, it is frequently used in robotic application for controlling DC motors.

There are two Enable pins on L293D. Pin 1 and pin 9, for being able to drive the motor, the pin 1 and 9 need to be high. For driving the motor with left H-bridge you need to enable pin 1 to high. And for right H-Bridge you need to make the pin 9 to high. If either pin1 or pin9 goes low then the motor in the corresponding section will suspend working.

**Working of L293D**

The 4 input pins for this L293D, pin 2, 7 on the left and pin 15, 10 on the right as shown on the pin diagram. Left input pins will regulate the rotation of motor connected on the left side and right input for motor on the right-hand side. The motors are rotated on the basis of the inputs provided at the input pins as LOGIC 1 or LOGIC 0.

In simple you need to provide Logic 0 or 1 across the input pins for rotating the motor.

**L293D Logic Table**

Let's consider a Motor connected on left side output pins (pin 3,6). For rotating the motor in clockwise direction, the input pins have to be given with Logic 1 and Logic 0.

- Pin 2 = Logic 1 and Pin 7 = Logic 0 | Clockwise Direction
- Pin 2 = Logic 0 and Pin 7 = Logic 1 | Anticlockwise Direction
- Pin 2 = Logic 0 and Pin 7 = Logic 0 | Idle [No rotation] [Hi-Impedance state]
- Pin 2 = Logic 1 and Pin 7 = Logic 1 | Idle [No rotation]

In a very similar way the motor can also operate across input pin 15,10 for motor on the right hand side.

**Voltage Specifications**

The voltage (Vcc) needed to for its own working is 5V but L293d will not use that Voltage to drive DC Motors. That means you should provide that voltage (36V maximum) and a maximum current of 600mA to drive the motors and maximum resistance is 60 ohms.

**Table 3:** Pin Description of L293D

| Pin No | Function | Name |
|---|---|---|
| 1 | Enable pin for Motor 1; active high | Enable 1,2 |
| 2 | Input 1 for Motor 1 | Input 1 |
| 3 | Output 1 for Motor 1 | Output 1 |
| 4 | Ground (0V) | Ground |
| 5 | Ground (0V) | Ground |
| 6 | Output 2 for Motor 1 | Output 2 |
| 7 | Input 2 for Motor 1 | Input 2 |
| 8 | Supply voltage for Motors; 9-12V (up to 36V) | $Vcc_2$ |
| 9 | Enable pin for Motor 2; active high | Enable 3,4 |
| 10 | Input 1 for Motor 1 | Input 3 |
| 11 | Output 1 for Motor 1 | Output 3 |
| 12 | Ground (0V) | Ground |
| 13 | Ground (0V) | Ground |
| 14 | Output 2 for Motor 1 | Output 4 |
| 15 | Input2 for Motor 1 | Input 4 |
| 16 | Supply voltage; 5V (up to 36V) | $Vcc_1$ |

### 4.3.6 LCD 2X16 Module

A **liquid-crystal display** (**LCD**) is a flat-panel display or other electronic visual display that uses the light-modulating properties of liquid crystals. Liquid crystals do not emit light directly. LCDs are used in a wide range of applications including computer monitors, televisions, instrument panels, aircraft cockpit displays, and signage. They are common in consumer devices such as DVD players, gaming devices, clocks, watches, calculators, and telephones, and have replaced cathode ray tube (CRT) displays in nearly all applications. They are available in a wider range of screen sizes than CRT and plasma displays, and since they do not use phosphors, they do not suffer image burn-in. LCDs are, however, susceptible to image persistence.

The LCD screen is more energy-efficient and can be disposed of more safely than a CRT. Its low electrical power consumption enables it to be used in battery-powered electronic equipment more efficiently than CRTs. It is an electronically modulated optical device made up of any number of segments controlling a layer of liquid crystals and arrayed in front of a light source (backlight) or reflector to produce images in color or monochrome.

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on.

A **16x2 LCD** means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

VEE pin is meant for adjusting the contrast of the LCD display and the contrast can be adjusted by varying the voltage at this pin. This is done by connecting one end of a POT to the Vcc (5V), other end to the Ground and connecting the center terminal (wiper) of of the POT to the VEE pin.

The JHD162A has two built in registers namely data register and command register. Data register is for placing the data to be displayed, and the command register is to place the commands. The 16×2 LCD module has a set of commands each meant for doing a particular job with the display.

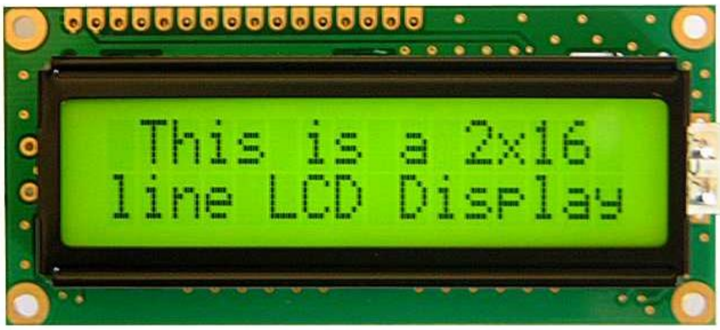



**Fig 13:** 2x16 Line LCD Display

**Table4:** Pin Description 2x16 Line LCD

| Pin No | Function | Name |
|---|---|---|
| 1 | Ground (0V) | Ground |
| 2 | Supply voltage; 5V (4.7V – 5.3V) | Vcc |
| 3 | Contrast adjustment; through a variable resistor | $V_{EE}$ |
| 4 | Selects command register when low; and data register when high | Register Select |
| 5 | Low to write to the register; High to read from the register | Read/write |
| 6 | Sends data to data pins when a high to low pulse is given | Enable |
| 7 | 8-bit data pins | DB0 |
| 8 | | DB1 |
| 9 | | DB2 |
| 10 | | DB3 |
| 11 | | DB4 |
| 12 | | DB5 |
| 13 | | DB6 |
| 14 | | DB7 |
| 15 | Backlight $V_{CC}$ (5V) | Led+ |
| 16 | Backlight Ground (0V) | Led- |

High logic at the RS pin will select the data register and Low logic at the RS pin will select the command register. If we make the RS pin high and the put a data in the 8-bit data line

(DB0 to DB7), the LCD module will recognize it as a data to be displayed. If we make RS pin low and put a data on the data line, the module will recognize it as a command.

R/W pin is meant for selecting between read and write modes. High level at this pin enables read mode and low level at this pin enables write mode E pin is for enabling the module. A high to low transition at this pin will enable the module.DB0 to DB7 are the data pins. The data to be displayed and the command instructions are placed on these pins.

LED+ is the anode of the back light LED and this pin must be connected to Vcc through a suitable series current limiting resistor. LED- is the cathode of the back light LED and this pin must be connected to ground.

## 16×2 LCD module commands.

16×2 LCD module has a set of preset command instructions. Each command will make the module to do a particular task. The commonly used commands and their function are given in the table below.

**Table 5:** 16×2 LCD module commands.

| **Command** | **Function** |
|---|---|
| 0F | LCD ON, Cursor ON, Cursor blinking ON |
| 01 | Clear screen |
| 02 | Return home |
| 04 | Decrement cursor |
| 06 | Increment cursor |
| 0E | Display ON ,Cursor blinking OFF |
| 80 | Force cursor to the beginning of 1$^{st}$ line |
| C0 | Force cursor to the beginning of 2$^{nd}$ line |
| 38 | Use 2 lines and 5×7 matrix |
| 83 | Cursor line 1 position 3 |
| 3C | Activate second line |
| 08 | Display OFF, Cursor OFF |
| C1 | Jump to second line, position1 |
| OC | Display ON, Cursor OFF |
| C1 | Jump to second line, position1 |
| C2 | Jump to second line, position2 |

**LCD initialization.**

The steps that have to be done for initializing the LCD display is given below and these steps are common for almost all applications.

- Send 38H to the 8-bit data line for initialization
- Send 0FH for making LCD ON, cursor ON and cursor blinking ON.
- Send 06H for incrementing cursor position.
- Send 01H for clearing the display and return the cursor.

**Sending data to the LCD.**

The steps for sending data to the LCD module is given below. I have already said that the LCD module has pins namely RS, R/W and E. It is the logic state of these pins that make the module to determine whether a given data input is a command or data to be displayed.

- Make R/W low.
- Make RS=0 if data byte is a command and make RS=1 if the data byte is a data to be displayed.
- Place data byte on the data register.
- Pulse E from high to low.
- Repeat above steps for sending another data.

## 4.3.7 Crystal Oscillator

An electronic circuit or electronic device that is used to generate periodically oscillating electronic signal is called as an electronic oscillator. The electronic signal produced by an oscillator is typically a sine wave or square wave. An electronic oscillator converts the direct current signal into an alternating current signal. The radio and television transmitters are broadcasted using the signals generated by oscillators. The electronic beep sounds and video game sounds are generated by the oscillator signals. These oscillators generate signals using the principle of oscillation.

There are different types of oscillator electronic circuits such as Linear oscillators – Hartley oscillator, Phase-shift oscillator, Armstrong oscillator, Clapp oscillator, Colpitts oscillator, and so on, Relaxation oscillators – Royer oscillator, Ring oscillator, Multivibrator, and so on, and Voltage Controlled Oscillator (VCO). In this article, let us discuss in detail about Crystal oscillator like what is crystal oscillator, crystal oscillator circuit, working, and use of crystal oscillator in electronic circuits.

**What is Crystal Oscillator?**

Quartz Crystal Oscillator

An electronic circuit that is used to generate an electrical signal of precise frequency by utilizing the vibrating crystal's mechanical resonance made of piezoelectric material. There are different types of piezoelectric resonators, but typically, quartz crystal is used in these types of oscillators. Hence, these oscillator electronic circuits are named as crystal oscillators.

**Fig 14:** Crystal Oscillator

**Crystal Oscillator Circuit Diagram**

The quartz crystal oscillator circuit diagram can be represented as follows:

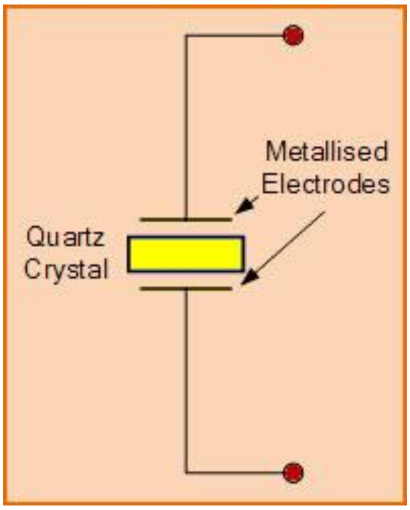



**Fig 15:** Crystal Oscillator Circuit Diagram

Electronic Symbol for Piezoelectric Crystal Resonator

The above diagram represents the electronic symbol for a piezoelectric crystal resonator which consists of two metalized electrodes and quartz crystal.

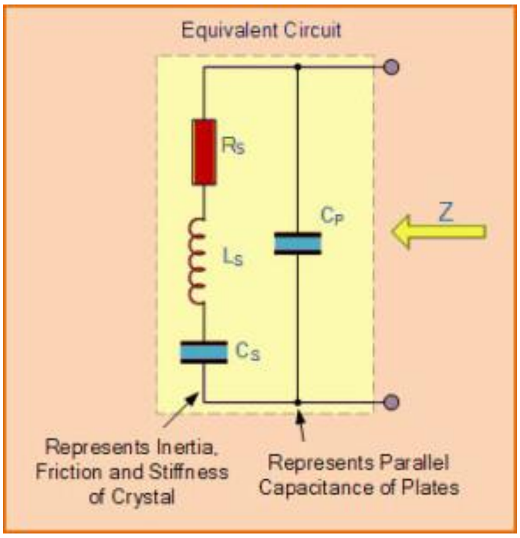



**Fig 16:** Equivalent Circuit Diagram of Quartz Crystal

The above figure shows the equivalent circuit diagram of quartz crystal in an electronic oscillator that consists of resistor, inductor, and capacitors which are connected as shown in the figure.

**Use of Crystal Oscillator**

- In general, we know that, crystal oscillators are used in the microprocessors and microcontrollers for providing the clock signals. Let us consider AVR Microcontroller for which an external crystal oscillator circuit of 12MHz is essential, even though (based on model) AVR requires 12 clock cycles for one machine cycle, such . that to give effective cycle rate at 1MHz (considering 12MHz clock) to 3.33MHz (considering maximum 16MHz clock). This crystal oscillator is used to generate clock pulses required for the synchronization of all the internal operations.
- There are numerous applications for crystal oscillator in various fields and a few crystal oscillator applications are shown below:
- **Application of Crystal Oscillator in Military and Aerospace**
- The use of crystal oscillator in military and aerospace, is to establish an efficient communication system, for the navigation purpose, electronic warfare, in the guidance systems, and so on.
- **Use of Crystal Oscillator in Research and Measurement**
- The crystal oscillator is used in research and measurement for celestrial navigation, space tracking purpose, in the measuring instruments and medical devices, and so on.
- **Industrial Applications of Crystal Oscillator**
- There are a huge number of industrial applications of crystal oscillator such as in computers, digital systems, instrumentation, phase locked loop systems, marine, modems, sensors, telecommunications, disk drives, and so on.
- **Use of Crystal Oscillator in Automotive**

- Crystal oscillator is used for engine controlling, stereo, clock and to trip computer, and in GPS system.
- **Consumer Applications of Crystal Oscillator**
- Crystal oscillators are used in many consumer goods such as cable television systems, personal computers, video cameras, toys and video games, radio systems, cellular phones, and so on.

### 4.3.8 Transformer

A **transformer** is an electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. Electromagnetic induction produces an electromotive force across a conductor which is exposed to time varying magnetic fields. Commonly, transformers are used to increase or decrease the voltages of alternating current in electric power applications.

**Fig 17:** Transformer

A varying current in the transformer's primary winding creates a varying magnetic flux in the transformer core and a varying magnetic field impinging on the transformer's secondary winding. This varying magnetic field at the secondary winding induces a varying electromotive force (EMF) or voltage in the secondary winding due to electromagnetic induction.

A Transformer takes in electricity at a higher voltage and lets it run through lots of coils wound around an iron core. ". A single-phase Transformer can operate to either increase or decrease the voltage applied to the primary winding. Because the current is alternating, the magnetism in the core is also alternating. Also around the core is an output wire with fewer coils. The magnetism changing back and forth makes a current in the wire. Having fewer coils means less voltage. When it is used to "decrease" the voltage on the secondary winding with respect to the primary it is called a **Step-down Transformer**. When a Transformer is used to "increase" the voltage on its secondary winding with respect to the primary, it is called a **Step-up Transformer**.

However, a third condition exists in which a transformer produces the same voltage on its secondary as is applied to its primary winding. In other words, its output is identical with respect to input. This type of Transformer is called an **"Impedance Transformer"** and is mainly used for impedance matching or the isolation of adjoining electrical circuits.

**Types**

Various specific electrical application designs require a variety of transformer types. Although they all share the basic characteristic transformer principles, they are customizing in construction or electrical properties for certain installation requirements or circuit conditions.

- *Autotransformer*: Transformer in which part of the winding is common to both primary and secondary circuits.[88]
- *Capacitor voltage transformer*: Transformer in which capacitor divider is used to reduce high voltage before application to the primary winding.
- *Distribution transformer, power transformer*: International standards make a distinction in terms of distribution transformers being used to distribute energy from transmission lines and networks for local consumption and power transformers being used to transfer electric energy between the generator and distribution primary circuits.[88][89][q]
- *Phase angle regulating transformer*: A specialised transformer used to control the flow of real power on three-phase electricity transmission networks.
- *Scott-T transformer*: Transformer used for phase transformation from three-phase to two-phase and vice versa.[88]

**Applications**

Transformers are used to increase (or step-up) voltage before transmitting electrical energy over long distances through wires. Wires have resistance which loses energy through joule heating at a rate corresponding to square of the current. By transforming power to higher voltage transformers enable economical transmission of power and distribution. Consequently, transformers have shaped the electricity supply industry, permitting generation to be located remotely from points of demand.[92] All but a tiny fraction of the world's electrical power has passed through a series of transformers by the time it reaches the consumer.[44]

Transformers are also used extensively in electronic products to decrease (or step-down) the supply voltage to a level suitable for the low voltage circuits they contain. The transformer also electrically isolates the end user from contact with the supply voltage.

Signal and audio transformers are used to couple stages of amplifiers and to match devices such as microphones and record players to the input of amplifiers. Audio transformers allowed telephone circuits to carry on a two-way conversation over a single pair of wires. A balun transformer converts a signal that is referenced to ground to a signal that has balanced voltages to ground, such

### 4.3.9 VOLTAGE REGULATORS

A **voltage regulator** is designed to automatically maintain a constant voltage level. A voltage regulator may be a simple "feed-forward" design or may include negative feedback control loops. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or DC voltages.

Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line.


**Fig 18:** Voltage Regulator

The **78xx** (sometimes **L78xx**, **LM78xx**, **MC78xx**...) is a family of self-contained fixed linear voltage regulator integrated circuits. The 78xx family is commonly used in electronic circuits requiring a regulated power supply due to their ease-of-use and low cost. For ICs within the family, the *xx* is replaced with two digits, indicating the output voltage (for example, the 7805 has a 5-volt output, while the 7812 produces 12 volts). The 78xx line are positive voltage regulators: they produce a voltage that is positive relative to a common ground. There is a related line of **79xx** devices which are complementary negative voltage regulators. 78xx and 79xx ICs can be used in combination to provide positive and negative supply voltages in the same circuit.

78xx ICs have three terminals and are commonly found in the TO-220 form factor, although they are available in surface-mount, TO-92, and TO-3 packages. These devices support an input voltage anywhere from around 2.5 volts over the intended output voltage up to a maximum of 35 to 40 volts depending on the model, and typically provide 1 or 1.5 amperes of current (though smaller or larger packages may have a lower or higher current rating)

## Advantages

- 78xx series ICs do not require additional components to provide a constant, regulated source of power, making them easy to use, as well as economical and efficient uses of space. Other voltage regulators may require additional components to set the output voltage level, or to assist in the regulation process. Some other designs (such as a switched-mode power supply) may need substantial engineering expertise to implement.
- 78xx series ICs have built-in protection against a circuit drawing too much current. They have protection against overheating and short-circuits, making them quite robust in most applications. In some cases, the current-limiting features of the 78xx devices can provide protection not only for the 78xx itself, but also for other parts of the circuit.

### 7805 Regulator

**7805** is a **voltage regulator** integrated circuit. It is a member of 78xx series of fixed linear voltage regulator ICs. The voltage source in a circuit may have fluctuations and would not give the fixed voltage output. The **voltage regulator IC** maintains the output voltage at a constant value.

The xx in 78xx indicates the fixed output voltage it is designed to provide. 7805 provides +5V regulated power supply. Capacitors of suitable values can be connected at input and output pins depending upon the respective voltage levels.

### Pin Description: 7805 voltage Regulator

| Pin No | Function | Name |
|---|---|---|
| 1 | Input voltage (5V-18V) | Input |
| 2 | Ground (0V) | Ground |
| 3 | Regulated output; 5V (4.8V-5.2V) | Output |

### 4.3.10 HC-05 Bluetooth Module

The HC-05 module is an easy-to-use Bluetooth SPP (Serial Port Protocol) module designed for transparent wireless serial communication between two devices. It behaves like a virtual serial cable, so any data sent from the microcontroller UART appears at the Android application side and vice versa without needing complex protocol handling in the user code. This makes it very convenient for projects where simple, reliable wireless data transfer is required.

The HC-05 is a fully qualified Bluetooth V2.0 + EDR (Enhanced Data Rate) module supporting up to 3 Mbps modulation with a complete 2.4 GHz radio transceiver and baseband integrated on the chip. It uses the CSR Bluecore 04-External single-chip Bluetooth system built on CMOS technology and supports Adaptive Frequency Hopping (AFH), which helps reduce interference from other 2.4 GHz devices and improves link reliability. With a compact footprint of about 12.7 mm × 27 mm and an onboard antenna, it occupies very little PCB area, simplifying mechanical design and integration into embedded systems.

Because the module provides a standard UART interface with configurable baud rate and simple AT command support, it significantly simplifies the overall design and development cycle for Bluetooth-enabled Arduino or microcontroller projects. Designers can quickly add wireless control or monitoring to their applications without needing to understand the low-level Bluetooth stack.

**Specifications**

**Fig 19:** HC-05 Bluetooth Module

**Hardware Features**

- Typical -80dBm sensitivity
- Up to +4dBm RF transmit power
- Low Power 1.8V Operation ,1.8 to 3.6V I/O
- PIO control
- UART interface with programmable baud rate
- With integrated antenna
- With edge connector

**Software Features**

- Default Baud rate: 38400, Data bits:8, Stop bit:1,Parity:No parity, Data control: has.

Supported baud rate: 9600,19200,38400,57600,115200,230400,460800.

- Given a rising pulse in PIO0, device will be disconnected.
- Status instruction port PIO1: low-disconnected, high-connected;
- PIO10 and PIO11 can be connected to red and blue led separately. When master and slave

are paired, red and blue led blinks 1time/2s in interval, while disconnected only blue led blinks 2times/s.

- Auto-connect to the last device on power as default.
- Permit pairing device to connect as default.
- Auto-pairing PINCODE:"0000" as default
- Auto-reconnect in 30 min when disconnected as a result of beyond the range of connection.

**AT command Default**

How to set the mode to server (master):

1. Connect PIO11 to high level.
2. Power on, module into command state.
3. Using baud rate 38400, sent the "AT+ROLE=1\r\n" to module, with "OK\r\n" means setting successes.
4. Connect the PIO11 to low level, repower the module, the module work as server (master).

**Bluetooth Master Mode:**

To configure the module as Bluetooth Master and to pair with another bluetooth module follow these steps. First, we need to put the module into command mode as above by pulling the CMD pin high before power on.

Enter these commands in order:

- AT+RMAAD Clear any paired devices
- AT+ROLE=1 Set mode to Master
- AT+RESET After changing role, reset is required
- AT+CMODE=0 Allow connection to any address
- AT+INQM=0,5,5 Inquire mode - Standard, stop after 5 devices found or after 5 seconds
- AT+PSWD=1234 Set PIN. Should be same as slave device
- AT+INIT Start Serial Port Profile (SPP) ( If Error(17) returned - ignore as profile already loaded)
- AT+INQ Start searching for devices

### 4.3.11 RESISTOR

A **resistor** is a passive two-terminal electrical component that implements electrical resistance as a circuit element. Resistors act to reduce current flow, and, at the same time, act to lower voltage levels within circuits. In electronic circuits, resistors are used to limit current flow, to adjust signal levels, bias active elements, and terminate transmission lines among other uses. High-power resistors, that can dissipate many watts of electrical power as heat, may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage. Variable resistors can be used to adjust circuit elements (such as a volume control or a lamp dimmer), or as sensing devices for heat, light, humidity, force, or chemical activity.

**Fig 20:** Resistor

### Types of Resistors

Resistors come in a variety of shapes and sizes. They might be through-hole or surface-mount. They might be a standard, static resistor, a pack of resistors, or a special variable resistor.

**Measurement**

The value of a resistor can be measured with an ohmmeter, which may be one function of a multimeter. Usually, probes on the ends of test leads connect to the resistor. A simple ohmmeter may apply a voltage from a battery across the unknown resistor (with an internal resistor of a known value in series) producing a current which drives a meter movement. The current, in accordance with Ohm's law, is inversely proportional to the sum of the internal resistance and the resistor being tested, resulting in an analog meter scale which is very non-linear, calibrated from infinity to 0 ohms.

A digital multimeter, using active electronics, may instead pass a specified current through the test resistance. The voltage generated across the test resistance in that case is linearly proportional to its resistance, which is measured and displayed. In either case the low-resistance ranges of the meter pass much more current through the test leads than do high-resistance ranges, in order for the voltages present to be at reasonable levels (generally below 10 volts) but still measurable.

**Termination and mounting**

Resistors will come in one of two termination-types: through-hole or surface-mount. These types of resistors are usually abbreviated as either PTH (plated through-hole) or SMD/SMT (surface-mount technology or device).

**Through-hole** resistors come with long, pliable leads which can be stuck into a breadboard or hand-soldered into a prototyping board or printed circuit board (PCB). These resistors are usually more useful in breadboarding, prototyping, or in any case where you'd rather not solder tiny, little 0.6mm-long SMD resistors. The long leads usually require trimming, and these resistors are bound to take up much more space than their surface-mount counterparts.

The most common through-hole resistors come in an axial package. The size of an axial resistor is relative to its power rating. A common ½W resistor measures about 9.2mm across, while a smaller ¼W resistor is about 6.3mm long.

**Surface-mount** resistors are usually tiny black rectangles, terminated on either side with even smaller, shiny, silver, conductive edges. These resistors are intended to sit on top of PCBs, where they're soldered onto mating landing pads. Because these resistors are so small, they're usually set into place by a robot, and sent through an oven where solder melts and holds them in place.

SMD resistors come in standardized sizes; usually either 0805 (0.8mm long by 0.5mm wide), 0603, or 0402. They're great for mass circuit-board-production, or in designs where space is a precious commodity. They take a steady, precise hand to manually solder, though!

**Applications of Resistors**

In electronic circuits, resistors play an important role to limit the current and provide only the required biasing to the vital active parts like the transistors and the ICs. We will try to find out what is the function of a resistor in electronics through the following illustrations:

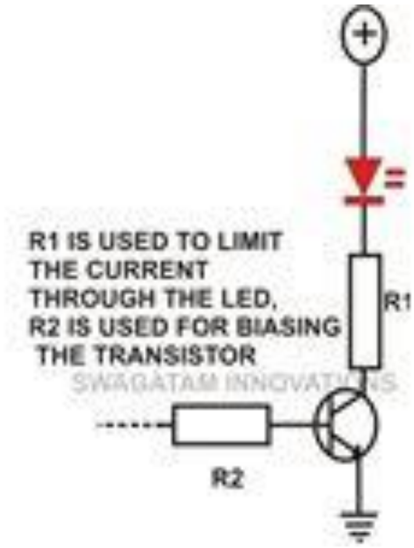



**Fig 20:** Resistor Application example

## 4.3.12 LED – Light Emitting Diode

A **light-emitting diode** (**LED**) is a two-lead semiconductor light source. It is a p–n junction diode, which emits light when activated.[4] When a suitable voltage is applied to the leads, electrons are able to recombine with electron holes within the device, releasing energy in the form of photons. This effect is called electroluminescence, and the color of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor.

An LED is often small in area (less than 1 mm$^2$) and integrated optical components may be used to shape its radiation pattern.[5]

Appearing as practical electronic components in 1962,[6] the earliest LEDs emitted low-intensity infrared light. Infrared LEDs are still frequently used as transmitting elements in remote-control circuits, such as those in remote controls for a wide variety of consumer electronics. The first visible-light LEDs were also of low intensity, and limited to red. Modern LEDs are available across the visible, ultraviolet, and infrared wavelengths, with very high brightness.

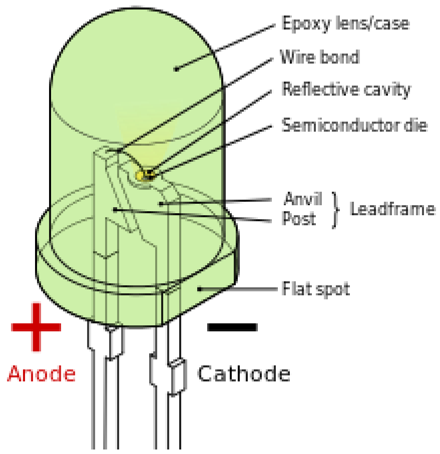



**Fig 21:** LED

Early LEDs were often used as indicator lamps for electronic devices, replacing small incandescent bulbs. They were soon packaged into numeric readouts in the form of seven-segment displays, and were commonly seen in digital clocks.

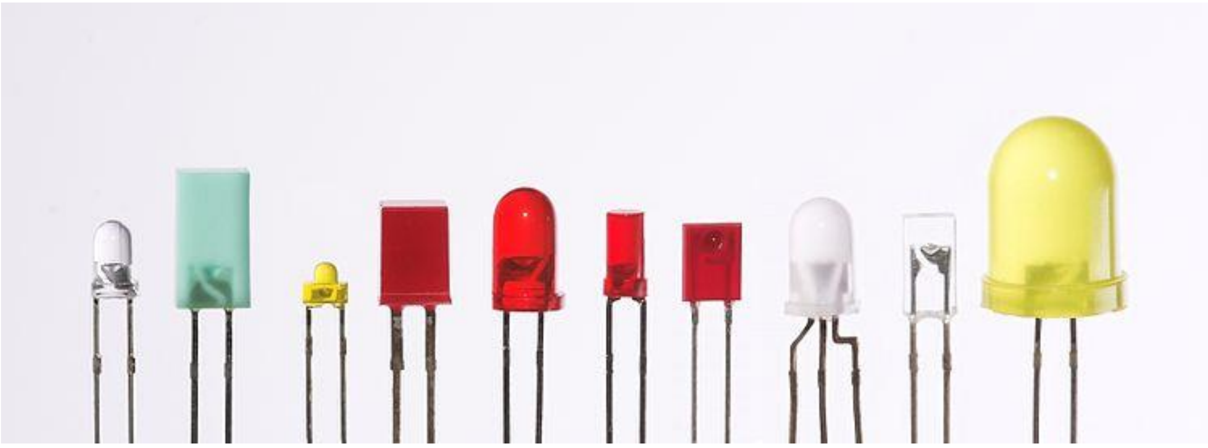
Recent developments in LEDs permit them to be used in environmental and task lighting. LEDs have many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved physical robustness, smaller size, and faster switching. Light-emitting diodes are now used in applications as diverse as aviation lighting, automotive headlamps, advertising, general lighting, traffic signals, camera flashes and lighted wallpaper. As of 2015, LEDs powerful enough for room lighting remain somewhat more expensive, and require more precise current and heat management, than compact fluorescent lamp sources of comparable output.

**Working principle**

A P-N junction can convert absorbed light energy into a proportional electric current. The same process is reversed here (i.e. the P-N junction emits light when electrical energy is applied to it). This phenomenon is generally called electroluminescence, which can be defined as the emission of light from a semi-conductor under the influence of an electric field. The charge carriers recombine in a forward-biased P-N junction as the electrons cross from the N-region and recombine with the holes existing in the P-region. Free electrons are in the conduction band of energy levels, while holes are in the valence energy band. Thus the energy level of the holes will be lesser than the energy levels of the electrons. Some portion of the energy must be dissipated in order to recombine the electrons and the holes. This energy is emitted in the form of heat and light.

The electrons dissipate energy in the form of heat for silicon and germanium diodes but in gallium arsenide phosphide (GaAsP) and gallium phosphide (GaP) semiconductors, the electrons dissipate energy by emitting photons. If the semiconductor is translucent, the junction becomes the source of light as it is emitted, thus becoming a light-emitting diode, but when the junction is reverse biased no light will be produced by the LED and, on the contrary, the device may also be damaged.

**Fig 22:** Types of LED



The main types of LEDs are miniature, high-power devices and custom designs such as alphanumeric or multi-colour

**Miniature**

These are mostly single-die LEDs used as indicators, and they come in various sizes from 2 mm to 8 mm, through-hole and surface mount packages. They usually do not use a separate heat sink. Typical current ratings range from around 1 mA to above 20 mA. The small size sets a natural upper boundary on power consumption due to heat caused by the high current density and need for a heat sink.

Common package shapes include round, with a domed or flat top, rectangular with a flat top (as used in bar-graph displays), and triangular or square with a flat top. The encapsulation may also be clear or tinted to improve contrast and viewing angle.

Researchers at the University of Washington have invented the thinnest LED. It is made of two-dimensional (2-D) flexible materials. It is three atoms thick, which is 10 to 20 times thinner than three-dimensional (3-D) LEDs and is also 10,000 times smaller than the thickness of a human hair. These 2-D LEDs are going to make it possible to create smaller, more energy-efficient lighting, optical communication and nano lasers.[118]

There are three main categories of miniature single die LEDs:

**Low-current**

Typically rated for 2 mA at around 2 V (approximately 4 mW consumption)

**Standard**

20 mA LEDs (ranging from approximately 40 mW to 90 mW) at around:

- 1.9 to 2.1 V for red, orange, yellow, and traditional green

- 3.0 to 3.4 V for pure green and blue
- 2.9 to 4.2 V for violet, pink, purple and white

**Ultra-high-output**

20 mA at approximately 2 or 4–5 V, designed for viewing in direct sunlight

5 V and 12 V LEDs are ordinary miniature LEDs that incorporate a suitable series resistor for direct connection to a 5 V or 12 V supply.

**Applications of LED**

LED uses fall into four major categories:

- Indicators and signs
- Lighting
- Data communication and other signaling
- Sustainable lighting
- Energy consumption
- Light sources for machine vision systems

## Lithium Ion Battery

Lithium-ion is the most popular rechargeable battery chemistry used today. Lithium-ion batteries power the devices we use every day, like our mobile phones and electric vehicles.

Lithium-ion batteries consist of single or multiple lithium-ion cells, along with a protective circuit board. They are referred to as batteries once the cell, or cells, are installed inside a device with the protective circuit board.

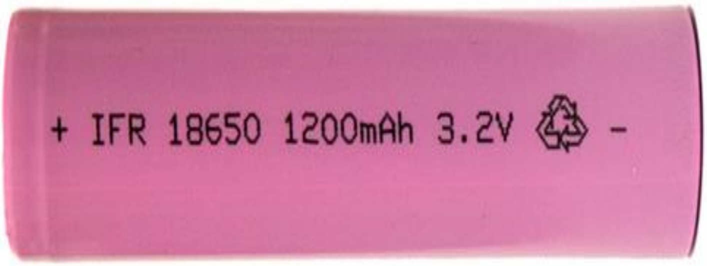



**Fig 23:** Lithium Ion Battery

- **Electrodes:** The positively and negatively charged ends of a cell. Attached to the current collectors
- **Anode:** The negative electrode
- **Cathode:** The positive electrode
- **Electrolyte:** A liquid or gel that conducts electricity

- **Current collectors:** Conductive foils at each electrode of the battery that are connected to the terminals of the cell. The cell terminals transmit the electric current between the battery, the device and the energy source that powers the battery
- **Separator:** A porous polymeric film that separates the electrodes while enabling the exchange of lithium ions from one side to the other

**How does a lithium-ion cell work?**

In a lithium-ion battery, lithium ions (Li+) move between the cathode and anode internally. Electrons move in the opposite direction in the external circuit. This migration is the reason the battery powers the device—because it creates the electrical current.
While the battery is discharging, the anode releases lithium ions to the cathode, generating a flow of electrons that helps to power the relevant device.
When the battery is charging, the opposite occurs: lithium ions are released by the cathode and received by the anode.

Its a 18650 1200mah Lithium Ion Battery. Its a 18650 Form Factor battery with 2900mah Capacity from BAK. It has a very high discharge rate of 3C. And has 1000 life cycles, which means that the battery has 80% capacity after 1000 cycles.

**Specifications**

- Nominal Capacity: 2900 maH
- Nominal Voltage: 3.7V
- Full Charge Voltage: 4.2V
- Discharge Rating: 3C
- Form Factor: 18650
- Life cycle: 1000 Cycles

# Chapter 5

# Implementation

## 5.1 Software Description and Implementation

## MIT App Inverter – Android Application Development Platform

**MIT App Inventor** is a web application integrated development environment originally provided by Google, and now maintained by the Massachusetts Institute of Technology (MIT). It allows newcomers to computer programming to create application software (apps) for two operating systems (OS): Android (operating system) |Android, and iOS.

It is free and open-source software released under Multi-licensing| dual licensing: a Creative Commons license# Attribution| Creative Commons Attribution Share Alike 3.0 Unported license, and an Apache License 2.0 for the source code.

It uses a graphical user interface (GUI) very similar to the programming languages Scratch (programming language) and the Star Logo, which allows users to drag and drop visual objects to create an application that can run on mobile devices. In creating App Inventor, Google drew upon significant prior research in educational computing, and work done within Google on online development environments.[1]

App Inventor and the projects on which it is based are informed by constructionist learning theories, which emphasize that programming can be a vehicle for engaging powerful ideas through active learning.

The MIT AI2 Companion app enables real-time debugging on connected devices via Wi-Fi, or Universal Serial Bus (USB).

### 5.1.2 SOFTWARE IMPLEMENTATION

After downloading, installing, and testing the Arduino software available at Arduino.cc (click on the Download link near the top of the page. This program is known as the Arduino IDE - short for Integrated Development Environment. Before you jump to the page for your operating system, make sure you’ve got all the right equipment. The programming environment outlined in this document is provided by Arduino.cc free of charge and is the recommended programming environment for your DuinoKit.

What you will need:

- A computer (Windows, Mac, or Linux)
- An Arduino-compatible microcontroller (DuinoKits use Arduino NANO w/ ATmega328 chip)
- A USB cable is required to connect the NANO microprocessor to your computer for programming

**Download the Arduino Software (IDE)**

Get the latest version from the download page. You can choose between the Installer (.exe) and the Zip packages. We suggest you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually. The Zip file is also useful if you want to create a portable installation.

When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.

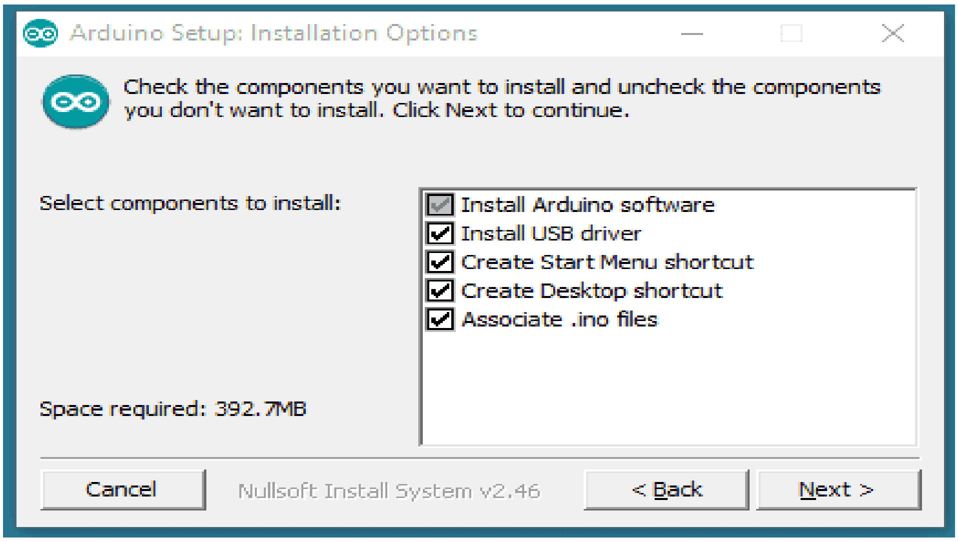



**Fig 24:** Choose the components to install.

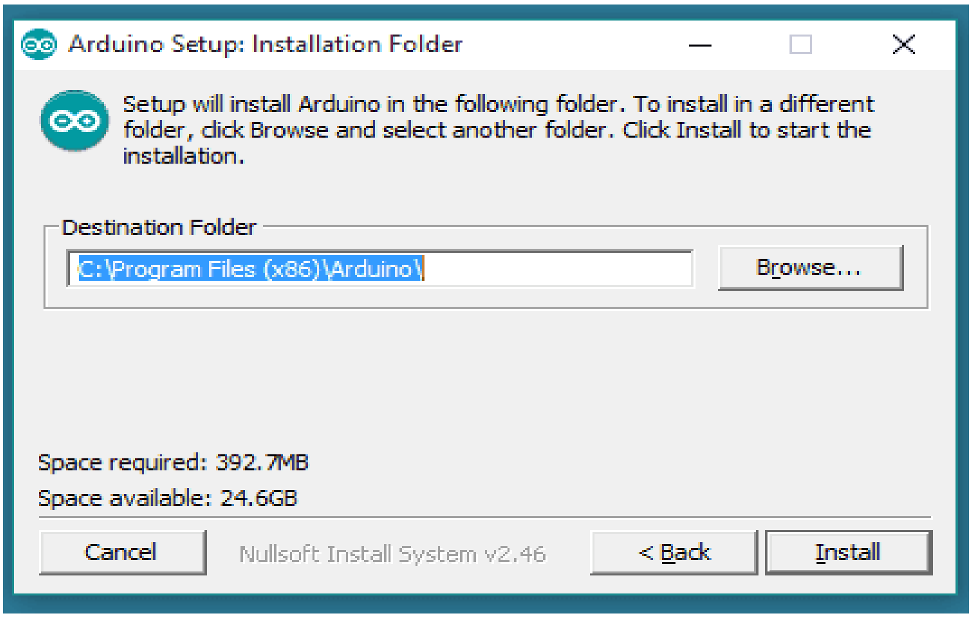



**Fig 25:** Choose the installation directory

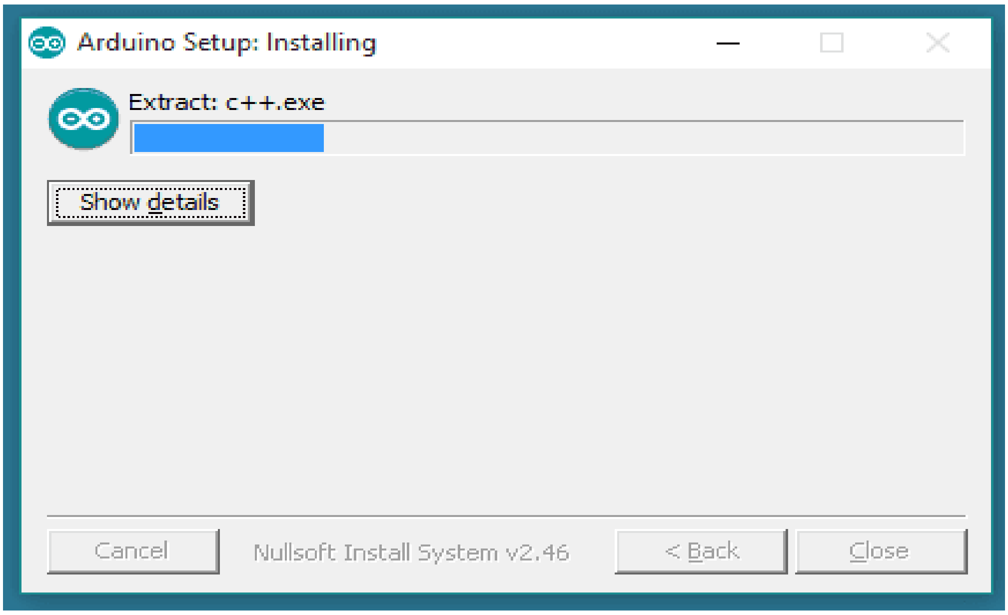


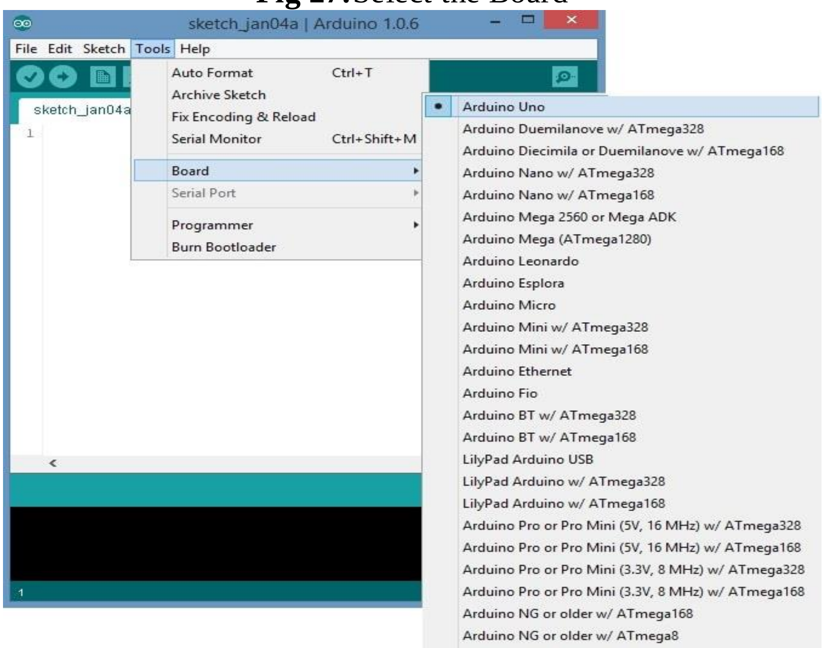

**Fig 26:** Installation in progress.

The process will extract and install all the required files to execute properly the Arduino Software (IDE). The text of the Arduino getting started guide is licensed under a Creative Commons Attribution-Share Alike 3.0 License. Code samples in the guide are released into the public domain

## CONFIGURING THE ARDUINO IDE

The next thing to do is to make sure the software is set up for your particular Arduino board. Go to the "Tools" drop-down menu, and find "Board". Another menu will appear, where you can select from a list of Arduino models. I have the Arduino Uno R3, so I chose "Arduino Uno".

**Fig 27:**Select the Board



## EXPLORING THE ARDUINO IDE

If you want, take a minute to browse through the different menus in the IDE. There is a good variety of example programs that come with the IDE in the "Examples" menu. These will help you get started with your Arduino right away without having to do lots of research:

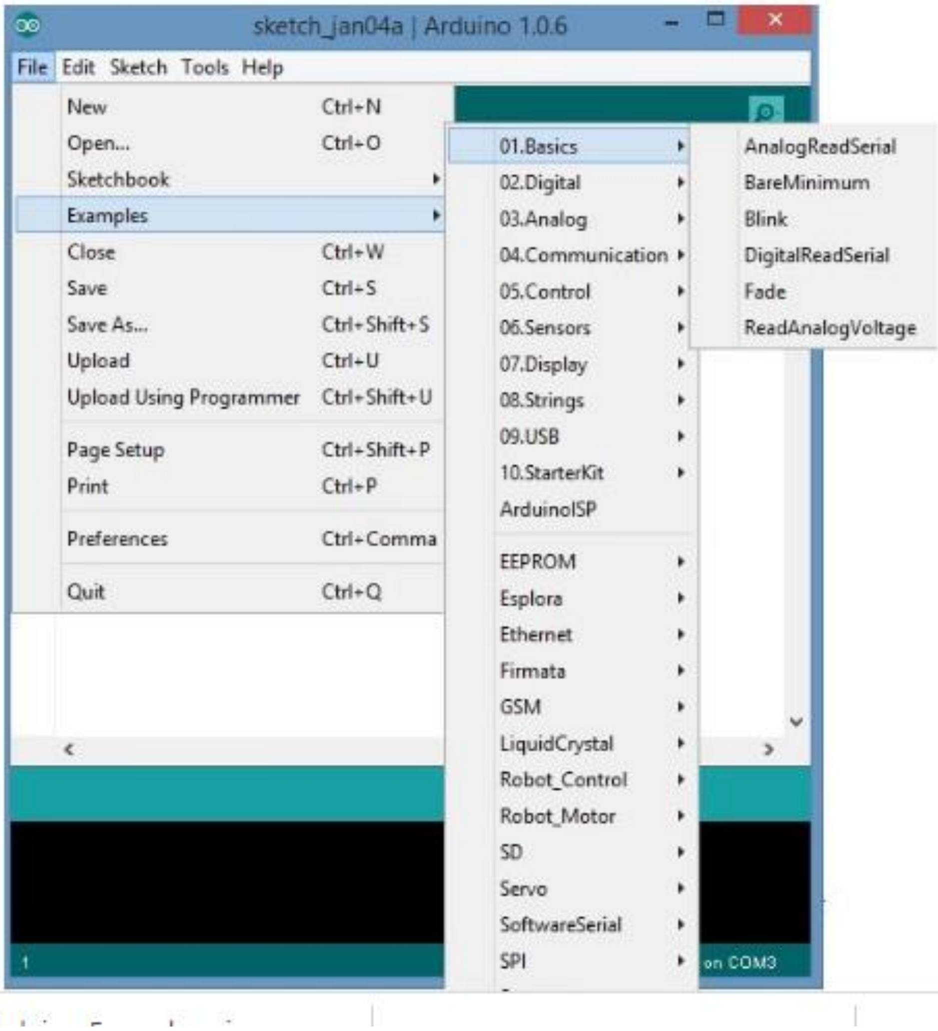



**Fig 28:**Example Programs in Library

## 5.2 Embedded System

An embedded system is a computer system—a combination of a computer processor, computer memory, and input/output peripheral devices—that has a dedicated function within a larger mechanical or electronic system.[1][2] It is *embedded* as part of a complete device often including electrical or electronic hardware and mechanical parts. Because an embedded system typically controls physical operations of the machine that it is embedded within, it often has real-time computing constraints. Embedded systems control many devices in common use today.[3] In 2009 it was estimated that ninety-eight percent of all microprocessors manufactured were used in embedded systems.[4][needs update]

Modern embedded systems are often based on microcontrollers (i.e. microprocessors with integrated memory and peripheral interfaces), but ordinary microprocessors (using external chips for memory and peripheral interface circuits) are also common, especially in more complex systems. In either case, the processor(s) used may be types ranging from general purpose to those specialized in a certain class of computations, or even custom designed for the application at hand. A common standard class of dedicated processors is the digital signal processor (DSP).

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Embedded systems range in size from portable personal devices such as digital watches and MP3 players to bigger machines like home appliances, industrial assembly lines, robots, transport vehicles, traffic light controllers, and medical imaging systems. Often they constitute subsystems of other machines like avionics in aircraft and spacecraft.

Large installations like factories, pipelines and electrical grids rely on multiple embedded systems networked together. Generalized through software customization, embedded systems such as programmable logic controllers frequently comprise their functional units.

Embedded systems range from those low in complexity, with a single microcontroller chip, to very high with multiple units, peripherals and networks, which may reside in equipment racks or across large geographical areas connected via long-distance communications lines.

**Development**

One of the first recognizably modern embedded systems was the Apollo Guidance Computer,[citation needed] developed ca. 1965 by Charles Stark Draper at the MIT Instrumentation Laboratory. At the project's inception, the Apollo guidance computer was considered the riskiest item in the Apollo project as it employed the then newly developed monolithic integrated circuits to reduce the computer's size and weight.

An early mass-produced embedded system was the Autonetics D-17 guidance computer for the Minuteman missile, released in 1961. When the Minuteman II went into production in 1966, the D-17 was replaced with a new computer that represented the first high-volume use of integrated circuits.

Since these early applications in the 1960s, embedded systems have come down in price and there has been a dramatic rise in processing power and functionality. An early microprocessor, the Intel 4004 (released in 1971), was designed for calculators and other small systems but still required external memory and support chips. By the early 1980s, memory, input and output system components had been integrated into the same chip as the processor forming a microcontroller. Microcontrollers find applications where a general-purpose computer would be too costly. As the cost of microprocessors and microcontrollers fell the prevalence of embedded systems increased.

Today, a comparatively low-cost microcontroller may be programmed to fulfill the same role as a large number of separate components. With microcontrollers, it became feasible to replace, even in consumer products, expensive knob-based analog components such as potentiometers and variable capacitors with up/down buttons or knobs read out by a microprocessor. Although in this context an embedded system is usually more complex than a traditional solution, most of the complexity is contained within the microcontroller itself. Very few additional components may be needed and most of the design effort is in the software. Software prototype and test can be quicker compared with the design and construction of a new circuit not using an embedded processor.

**Applications of Embedded System**

Embedded systems are commonly found in consumer, industrial, automotive, home appliances, medical, telecommunication, commercial, and aerospace and military applications.

Telecommunications systems employ numerous embedded systems from telephone switches for the network to cell phones at the end user. Computer networking uses dedicated routers and network bridges to route data.

Consumer electronics include MP3 players, television sets, mobile phones, video game consoles, digital cameras, GPS receivers, and printers. Household appliances, such as microwave ovens, washing machines and dishwashers, include embedded systems to provide flexibility, efficiency and features. Advanced HVAC systems use networked thermostats to more accurately and efficiently control temperature that can change by time of day and season. Home automation uses wired- and wireless-networking that can be used to control lights, climate, security, audio/visual, surveillance, etc., all of which use embedded devices for sensing and controlling.

Transportation systems from flight to automobiles increasingly use embedded systems. New airplanes contain advanced avionics such as inertial guidance systems and GPS receivers that also have considerable safety requirements. Spacecraft rely on avionics systems for trajectory correction. Various electric motors — brushless DC motors, induction motors and DC motors — use electronic motor controllers. Automobiles, electric vehicles, and hybrid vehicles increasingly use embedded systems to maximize efficiency and reduce pollution. Other automotive safety systems using embedded systems include anti-lock braking system (ABS), Electronic Stability Control (ESC/ESP), traction control (TCS) and automatic four-wheel drive.

Medical equipment uses embedded systems for monitoring, and various medical imaging (PET, Single-photon emission computed tomography (SPECT), CT, and MRI) for non-invasive internal inspections. Embedded systems within medical equipment are often powered by industrial computers.[8]

Embedded systems are used for safety-critical systems in aerospace and defense industries. Unless connected to wired or wireless networks via on-chip 3G cellular or other methods for IoT monitoring and control purposes, these systems can be isolated from hacking and thus be more secure.[citation needed] For fire safety, the systems can be designed to have a greater ability to handle higher temperatures and continue to operate. In dealing with security, the embedded systems can be self-sufficient and be able to deal with cut electrical and communication systems.

Miniature wireless devices called motes are networked wireless sensors. Wireless sensor networking makes use of miniaturization made possible by advanced IC design to couple full wireless subsystems to sophisticated sensors, enabling people and companies to measure a myriad of things in the physical world and act on this information through monitoring and control systems. These motes are completely self-contained and will typically run off a battery source for years before the batteries need to be changed or charged.

## 5.3 Embedded C Programming

**Embedded C** is a set of language extensions for the C programming language by the C Standards Committee to address commonality issues that exist between C extensions for different embedded systems.

Embedded C programming typically requires nonstandard extensions to the C language in order to support enhanced microprocessor features such as fixed-point arithmetic, multiple distinct memory banks, and basic I/O operations. The C Standards Committee produced a Technical Report, most recently revised in 2008[1] and reviewed in 2013,[2] providing a common standard for all implementations to adhere to. It includes a number of features not available in normal C, such as fixed-point arithmetic, named address spaces and basic I/O hardware addressing. Embedded C uses most of the syntax and semantics of standard C, e.g., main() function, variable definition, datatype declaration, conditional statements (if, switch case), loops (while, for), functions, arrays and strings, structures and union, bit operations, macros, etc.[3]

Embedded C is most popular programming language in software field for developing electronic gadgets. Each processor used in electronic system is associated with embedded software.

Embedded C programming plays a key role in performing specific function by the processor. In day-to-day life we used many electronic devices such as mobile phone, washing machine, digital camera, etc. These all device working is based on microcontroller that are programmed by embedded C.

Let's see the block diagram representation of embedded system programming:

The Embedded C code written in above block diagram is used for blinking the LED connected with Port0 of microcontroller.

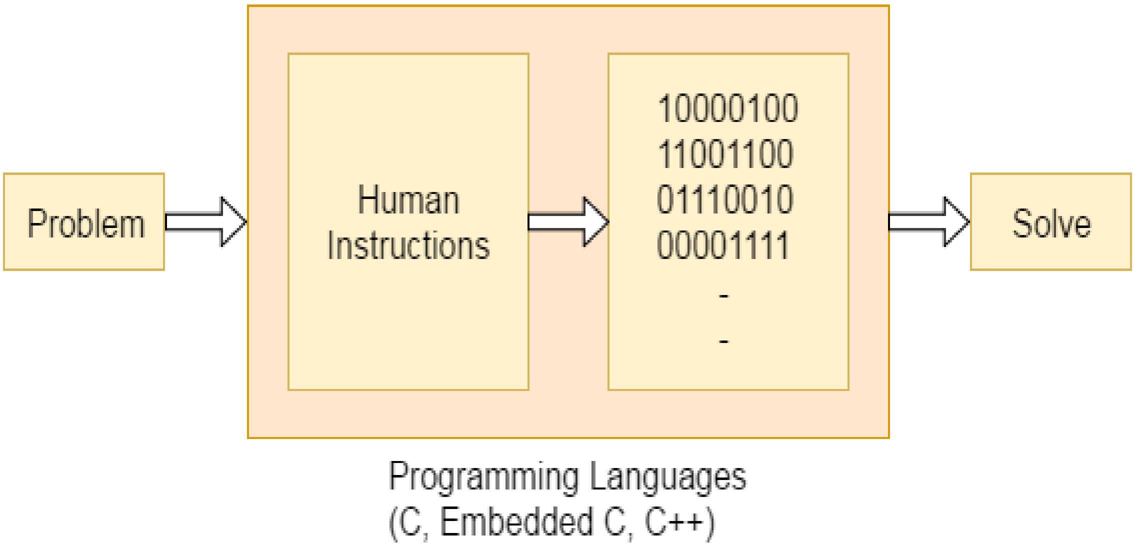
In embedded system programming C code is preferred over other language. Due to the following reasons:

- Easy to understand
- High Reliability
- Portability
- Scalability

**Basic Declaration**

Let's see the block diagram of Embedded C Programming development:

**Fig 29:** C Programming development




Function is a collection of statements that is used for performing a specific task and a collection of one or more functions is called a programming language. Every language is consisting of basic elements and grammatical rules. The C language programming is designed for function with variables, character set, data types, keywords, expression and so on are used for writing a C program.

The extension in C language is known as embedded C programming language. As compared to above the embedded programming in C is also have some additional features like data types, keywords and header file etc is represented by

**Basic Embedded C Programming Steps**

- Let's see the block diagram representation of Embedded C Programming Steps:
-
- The microcontroller programming is different for each type of operating system. Even though there are many operating system are exist such as Windows, Linux, RTOS, etc but RTOS has several advantages for embedded system development.

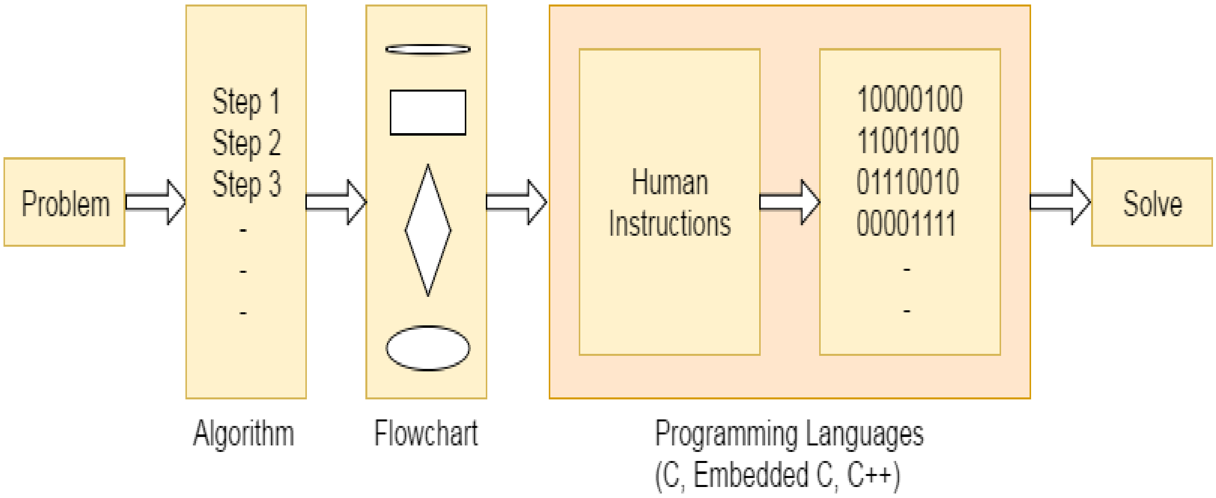



**Fig 30:** Embedded C Programming Steps

# CHAPTER 6
# Results and Discussion

## 6.1 System Working Screens / Photos

**Fig 31:** Android Application made By MIT Software

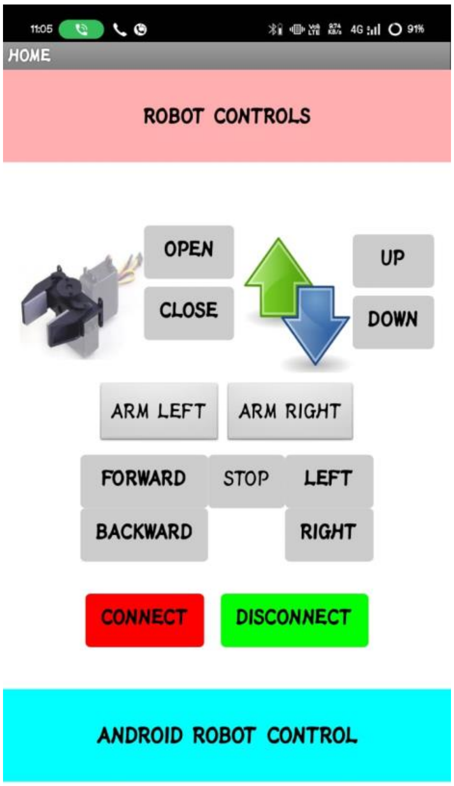



**Fig 32:** LCD Display

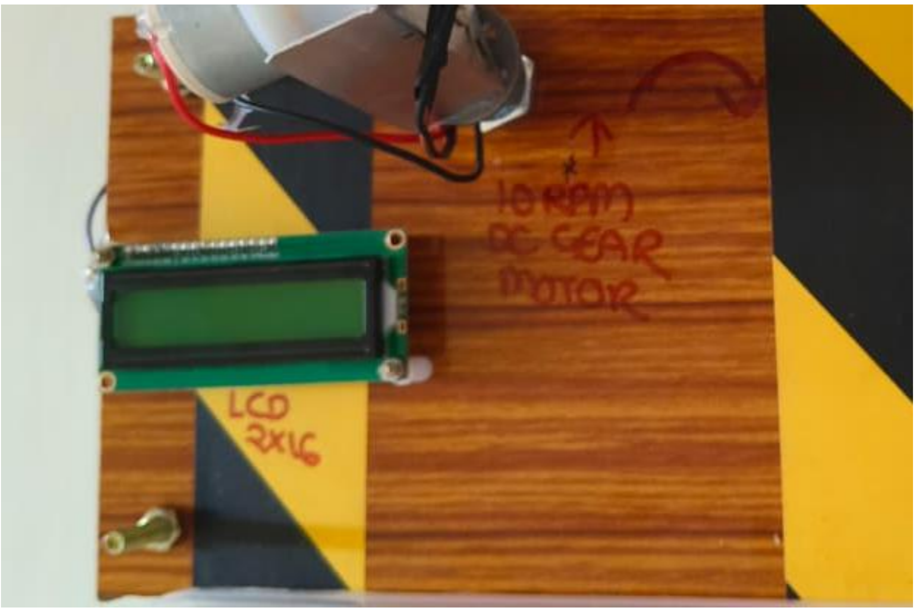


**Fig 33:** Robotic Arm

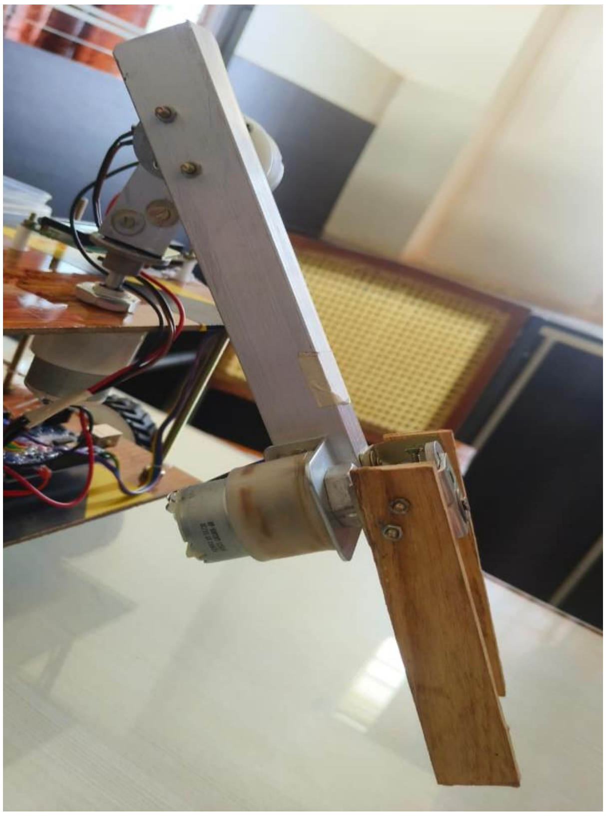


**Fig 35:** Battery /Power Source and Load Container

**Fig 34:** Circuit components Arrangement

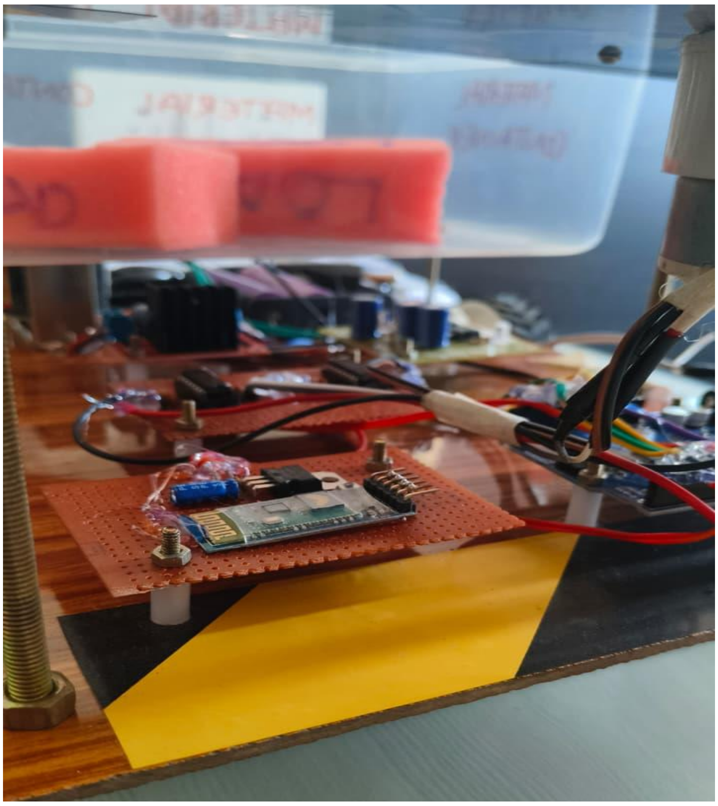
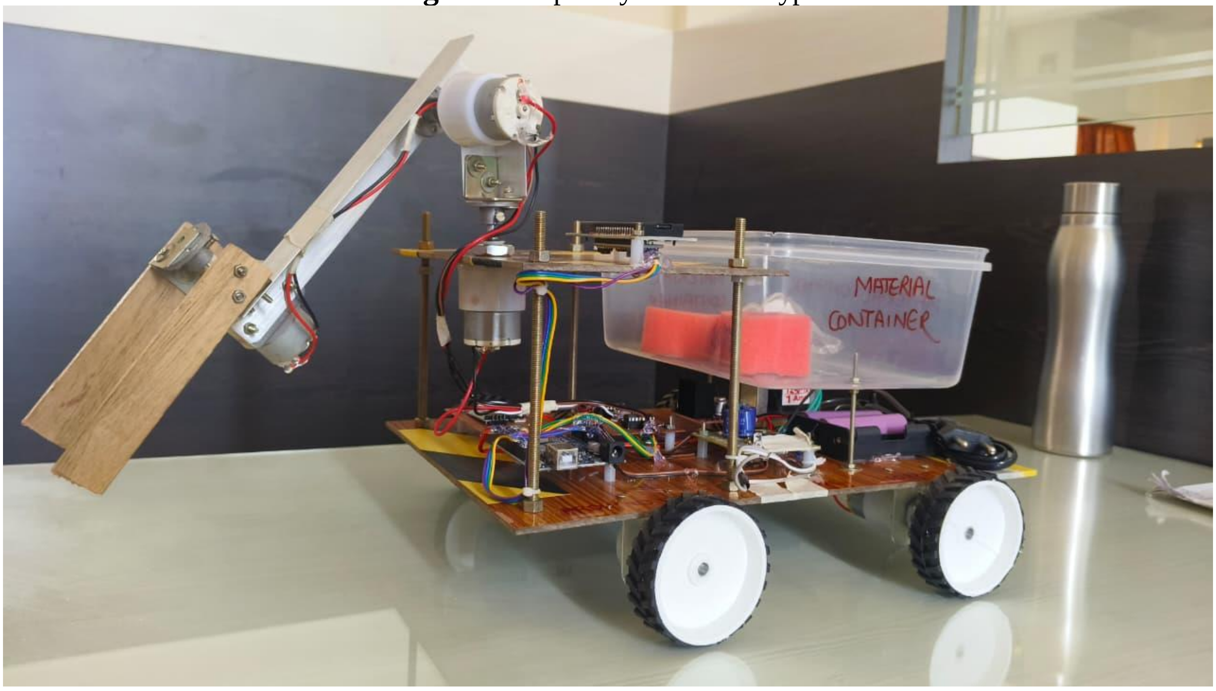


**Fig 36:** Completely Built Prototype




**ANDROID CONTROLLED PICK AND PLACE ROBOTIC ARM VEHICLE**

## 6.2 Observations and Performance

The Android application successfully established a Bluetooth connection with the robot, and the “Bluetooth Connected” message was reliably shown on the LCD, confirming proper pairing and communication.

Using the UP, DOWN, LEFT, RIGHT, OPEN and CLOSE controls, the robotic vehicle could move forward, backward, left and right, and the gripper could open and close without noticeable delay, showing that the command-response time is acceptable for small-scale handling.

The 12 V battery along with the 7805-based regulated power supply provided stable operation for the Arduino, motor drivers and Bluetooth module, and no abnormal reset or brown-out behaviour was observed during continuous movement and pick-and-place actions. The robot was able to pick, carry and place small objects of suitable weight accurately within its working range, demonstrating that the geared motors and mechanical design are adequate for educational and light industrial prototype applications.

The LCD status messages during connection, command execution and disconnection helped the user understand the current state of the system, which improves usability and makes debugging easier if any malfunction occurs.
Overall, the proposed system practically overcomes major limitations of the existing systems such as wired control, lack of portability and high cost, but its payload and working area are still limited, indicating scope for future improvement in mechanical strength and range.

## 6.3 Advantages of Industrial ARM

**Quality**

Industrial robotic arms can perform the same motion with very high precision and repeatability, cycle after cycle. Each weld, placement or assembly operation follows the programmed path and parameters exactly, which reduces variation between parts and improves overall product consistency. Because robots do not suffer from fatigue or loss of concentration, defects caused by human error are greatly reduced, leading to higher yield and less rework or rejection.

**Production**

By automating tasks with an industrial arm, the production line can run at higher and more uniform speeds compared to manual operation. Robots can work continuously for long shifts without breaks, so the number of parts handled per hour increases significantly. This higher throughput directly raises total output and helps meet tight deadlines, and robots can also be reprogrammed to handle different products or batches with minimal changeover time, improving flexibility in production planning.

**Safety**

Industrial robotic arms take over tasks that are risky, dirty, or ergonomically harmful for human workers, such as handling heavy loads, working near hot surfaces, or using hazardous tools and chemicals. By moving operators from direct manual handling to supervisory and monitoring roles, the chances of accidents, injuries and long-term health issues are reduced. As a result, the working environment becomes safer, and companies can comply more easily with safety standards while still maintaining or increasing productivity.

# Chapter 7
# Conclusion and Future Scope

## 7.1 Conclusion

Building the industrial arm control system was successfully achieved using the Arduino microcontroller board as the main controller. This industrial ARM provides 3 degrees of freedom, enabling basic industrial movements required for pick and place applications. Using 10 RPM DC geared motors, the complete motion of the arm joints and base is controlled smoothly to handle small objects effectively. The motors are driven through the L293 motor driver IC, which interfaces the low-power control signals from the Arduino with the higher current required by the motors.

To operate the system wirelessly, an Android application is used as the user interface for sending movement and gripper commands to the ARM. This application is developed using the user-friendly online platform App Inventor (www.appinventor.mit.edu), which simplifies the process of creating custom Android apps without complex coding. Overall, the project demonstrates a low-cost, flexible and educational solution for understanding industrial robotic arm control using embedded systems and mobile-based wireless operation.

## 7.2 Future Scope

The proposed Android-based pick and place robot provides a strong foundation for further enhancements in automation and intelligent systems. Future improvements can significantly increase its efficiency, intelligence, and industrial applicability.

**1. Integration of Computer Vision**

- Add a camera module (ESP32-CAM or USB camera)
- Use image processing or AI (TensorFlow Lite, OpenCV)
- Enable automatic object detection, color sorting, and position tracking

**2. IoT-Based Remote Monitoring and Control**

- Upgrade from Bluetooth to Wi-Fi/IoT platforms (Blynk, MQTT, Firebase)
- Control and monitor the robot from anywhere via the internet
- Real-time data logging and performance tracking

**3. Autonomous Operation**

- Implement sensors (IR, Ultrasonic) for obstacle detection
- Enable automatic navigation and path planning
- Reduce dependency on manual mobile control

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- Internet Source https://www.elprocus.com/pick-nplace-robot/ Jong Hoon Ahnn, “The Robot control using the wireless communication and the serial communication.” May 2007. A.S.C.S. Sastry
- K.N.H. Srinivas (2010), “An automated microcontroller based the mixing system”, International journal on computer science and engineering. (Volume II, Issue 8, August 2010).
- B.O.Omijeh, R. Uhunmwangho, M. Ehikhamenle, “Design analysis of a remote-controlled pick and place robotic vehicle”, International journal of Engineering Research and Development. (Volume 10, Issue 5, May 2014).

# APPENDIX

**Appendix:**

```
#include <Servo.h>


#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);


#include <SoftwareSerial.h>
SoftwareSerial myserial (2, 3);


int L1 = A0, L2 = A1, R1 = A2, R2 = A3;


void setup()
{
  Serial.begin(9600);
  myserial.begin(9600);
  delay(100);


  pinMode(R1, OUTPUT);
  pinMode(R2, OUTPUT);
  pinMode(L1, OUTPUT);
  pinMode(L2, OUTPUT);
```

```
digitalWrite(R1,LOW); // ROBOT

digitalWrite(R2,LOW);

digitalWrite(L1,LOW); //

digitalWrite(L2,LOW);


pinMode(4, OUTPUT);

pinMode(5, OUTPUT);

pinMode(6, OUTPUT);

pinMode(7, OUTPUT);

pinMode(8, OUTPUT);

pinMode(9, OUTPUT);


digitalWrite(4,LOW); // ARM

digitalWrite(5,LOW);

digitalWrite(6,LOW); //

digitalWrite(7,LOW);

digitalWrite(8,LOW); //

digitalWrite(9,LOW);


lcd.init();

lcd.backlight();


lcd.clear();

lcd.setCursor(0, 0);

lcd.print("    ANDROID    ");
```

```
  lcd.setCursor(0, 1);
  lcd.print("     BASED     ");
  delay(1000);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(" PICK AND PLACE ");
  lcd.setCursor(0, 1);
  lcd.print("     ROBOT     ");
  delay(1000);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);

}

void loop()
{

  if (myserial.available() > 0)
  {
```

```
char c = myserial.read();
Serial.print("INPUT : ");
Serial.println(c);

if (c == 'F')
{
  //Serial.println("FORWARD");
  f();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("FORWARD");
  delay(400);
}

else if (c == 'B')
{
  //Serial.println("BACKWARD");
  b();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("BACKWARD");
```

```
  delay(400);
  s();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}

else if (c == 'L')
{
  Serial.println("LEFT");
  l();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("LEFT");
  delay(500);
  s();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
```

```
  lcd.print("STOP");
  delay(500);
}


else if (c == 'R')
{
  Serial.println("RIGHT");
  r();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("RIGHT");
  delay(500);
  s();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}
else if(c == 'S')
{
  s();
```

```
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}

/********* ARM ***********/

else if (c == 'O')
{
  //Serial.println("OPEN");

  digitalWrite(7,HIGH); // ARM
  digitalWrite(6,LOW);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("GRIPPER OPEN");
  delay(100);

  digitalWrite(6,LOW); // ARM
```

```
  digitalWrite(7,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}


else if (c == 'C')
{
  //Serial.println("CLOSE");


  digitalWrite(6,HIGH); // ARM
  digitalWrite(7,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("CUTTER CLOSE");
  delay(100);


  digitalWrite(6,LOW); // ARM
```

```
  digitalWrite(7,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}


else if (c == 'U')
{
  Serial.println("UP");


  digitalWrite(4,HIGH); //
  digitalWrite(5,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("ARM UP");
  delay(200);


  digitalWrite(4,LOW); //
```

```
  digitalWrite(5,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}

else if (c == 'D')
{
  Serial.println("DOWN");


  digitalWrite(4,LOW); //
  digitalWrite(5,HIGH);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("ARM DOWN");
  delay(200);


  digitalWrite(4,LOW); //
```

```
  digitalWrite(5,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}


 else if (c == 'X')
 {
  Serial.println("ALEFT");


  digitalWrite(8,HIGH); //
  digitalWrite(9,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("ARM LEFT");
  delay(200);


  digitalWrite(8,LOW); //
```

```
  digitalWrite(9,LOW);


  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("STOP");
  delay(500);
}

else if (c == 'Y')
{
  Serial.println("ARIGHT");

  digitalWrite(8,LOW); //
  digitalWrite(9,HIGH);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("ROBOT STATUS");
  lcd.setCursor(0, 1);
  lcd.print("ARM RIGHT");
  delay(200);

  digitalWrite(8,LOW); //
```

```
      digitalWrite(9,LOW);


      lcd.clear();

      lcd.setCursor(0, 0);

      lcd.print("ROBOT STATUS");

      lcd.setCursor(0, 1);

      lcd.print("STOP");

      delay(500);

    }

  }

}

// END OF LOOP


void f()

{

  digitalWrite(L2, HIGH);

  digitalWrite(L1, LOW);

  digitalWrite(R2, HIGH);

  digitalWrite(R1, LOW);




}

void b()

{

  digitalWrite(L1, HIGH);
```

```
  digitalWrite(L2, LOW);

  digitalWrite(R1, HIGH);

  digitalWrite(R2, LOW);


}


void l()

{

    digitalWrite(L1, LOW);

  digitalWrite(L2, HIGH);

  digitalWrite(R1, HIGH);

  digitalWrite(R2, LOW);

}


void r()

{


  digitalWrite(L1, HIGH);

  digitalWrite(L2, LOW);

  digitalWrite(R1, LOW);

  digitalWrite(R2, HIGH);

}


void s()

{
```

```
  digitalWrite(L1, LOW);

  digitalWrite(L2, LOW);

  digitalWrite(R1, LOW);

  digitalWrite(R2, LOW);

}
```

## STUDENTDETAILS

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